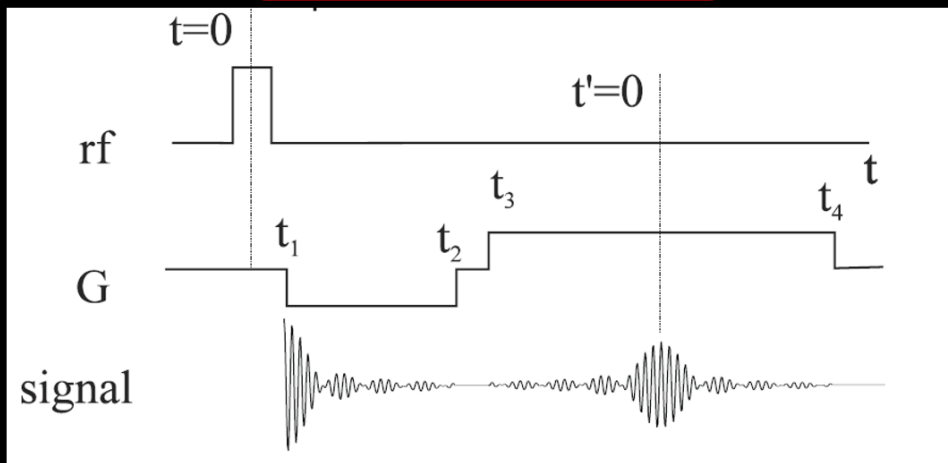
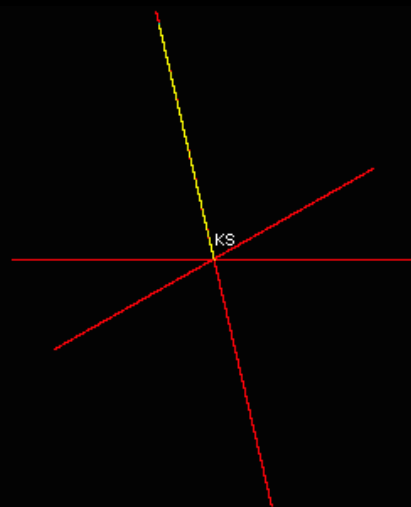
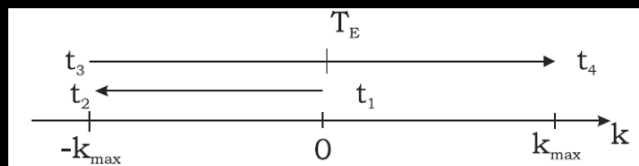


Short Recap

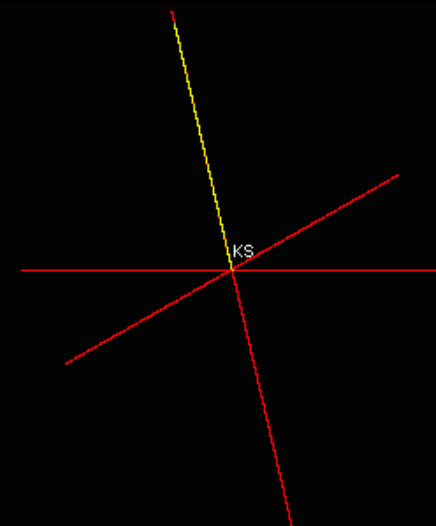
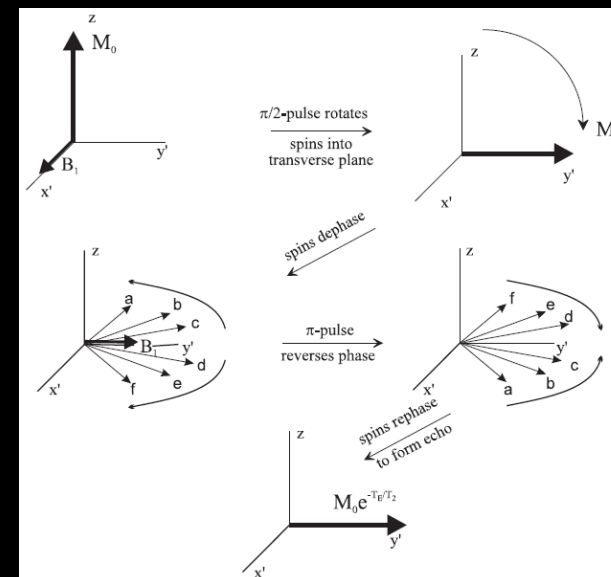
Gradient Echo



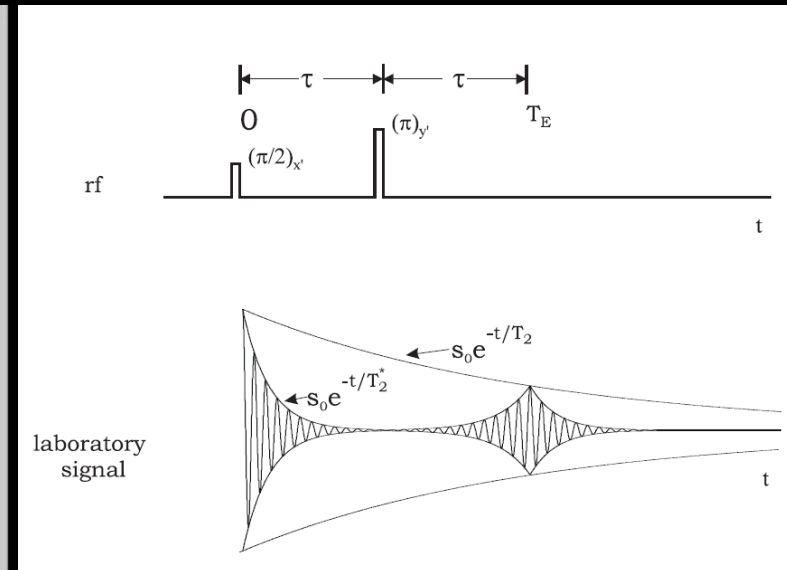
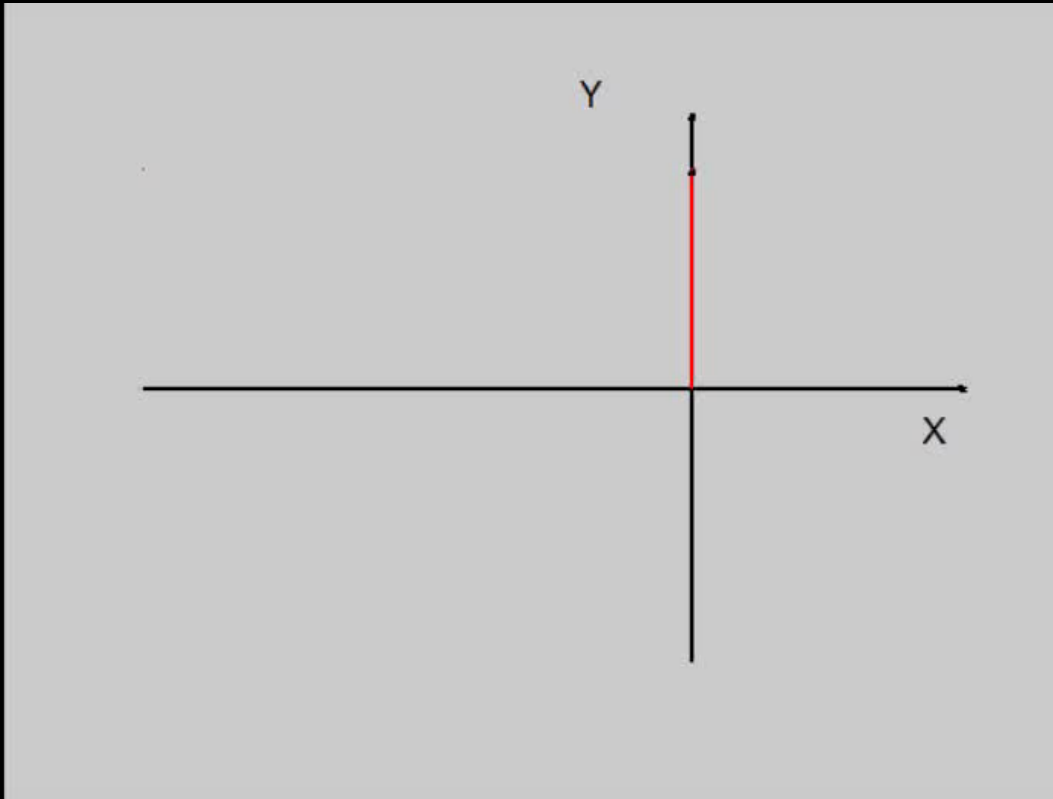
$$k_x = \gamma \cdot G_x \cdot t$$



Spin Echo (rf echo)



Spin echo envelope



Signal decay
 T_2^* vs T_2

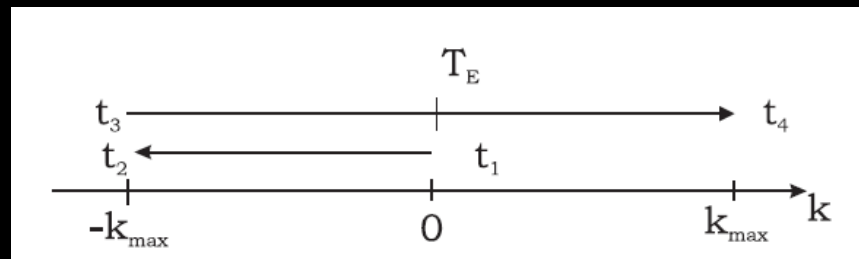
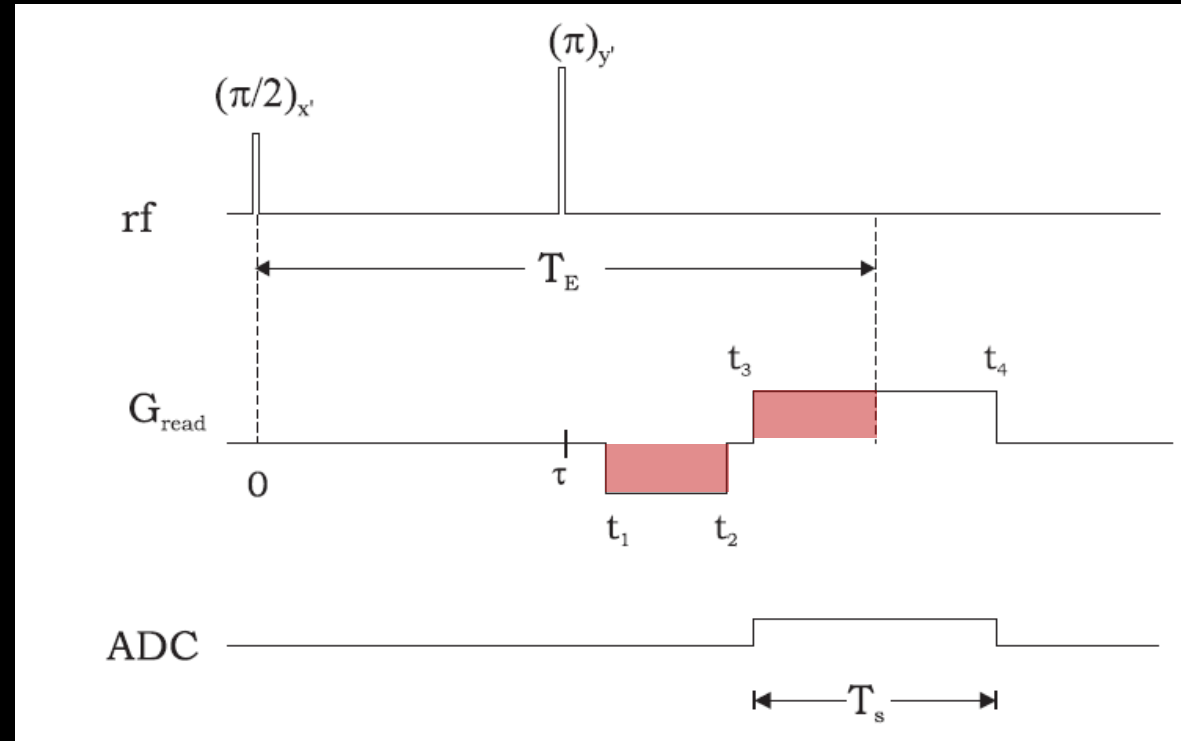
Note that signal decay in the presence of a gradient is faster than, otherwise.

You do not need a gradient to get a spin echo

... however, you need a gradient to encode spatial information (i.e. to generate an image)

Spin echo

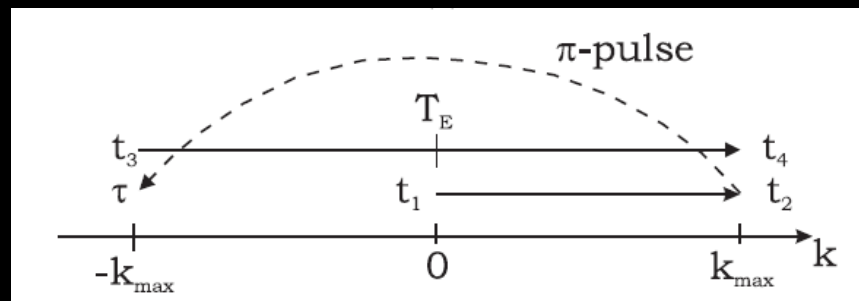
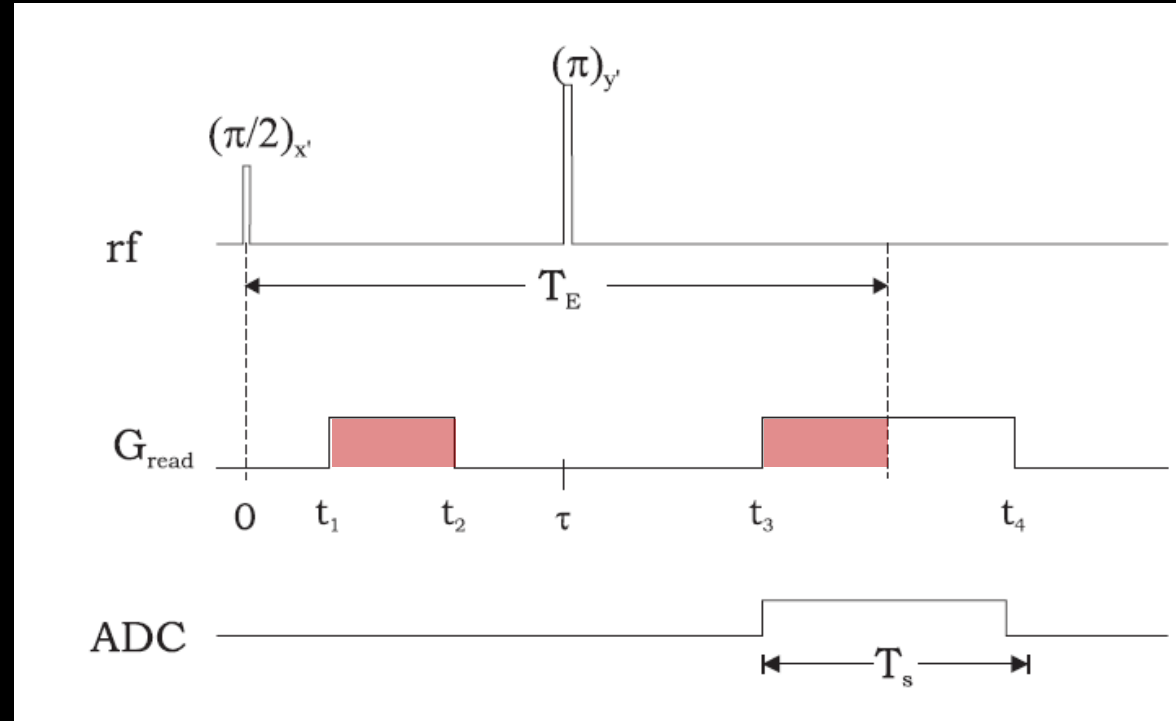
W/Spatial encoding



Spin echo

W/Spatial encoding

...alternatively



Multidimensional Spatial Encoding (2D/3D imaging)

$$1\text{D: } s(k_x) = \int dx \cdot \rho(x) \cdot e^{-i \cdot 2 \cdot \pi \cdot k \cdot x}$$



$$x \rightarrow \vec{r} = (x, y, z)$$

$$3\text{D: } s(k) = \int d^3r \cdot \rho(r) \cdot e^{-i2\pi(\vec{r} \cdot \vec{k})}$$

- 1) All dimensions are equivalent
- 2) k along each direction is determined by the gradient in that direction

$$\begin{aligned} & s(k_x, k_y, k_z) \\ &= \iiint dx \cdot dy \cdot dz \cdot \rho(x, y, z) \cdot e^{-i2\pi(k_x x + k_y y + k_z z)} \end{aligned}$$

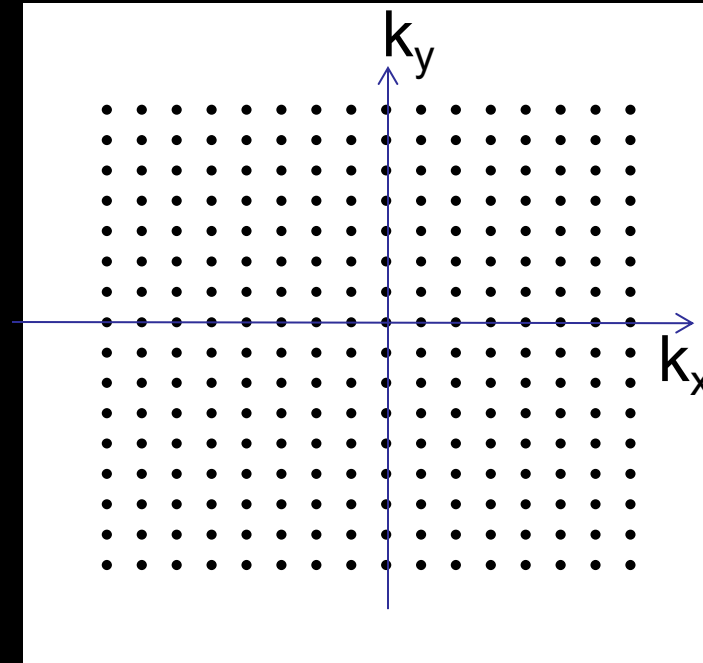
Multidimensional Spatial Encoding (2D/3D imaging)

Practically

- The goal:
Collect every point in the 3 dimensional k-space

- The means
Use gradients to traverse the k-space

$$k_i(t) = \gamma \cdot G_i \cdot i \cdot t, i = x, y, z$$

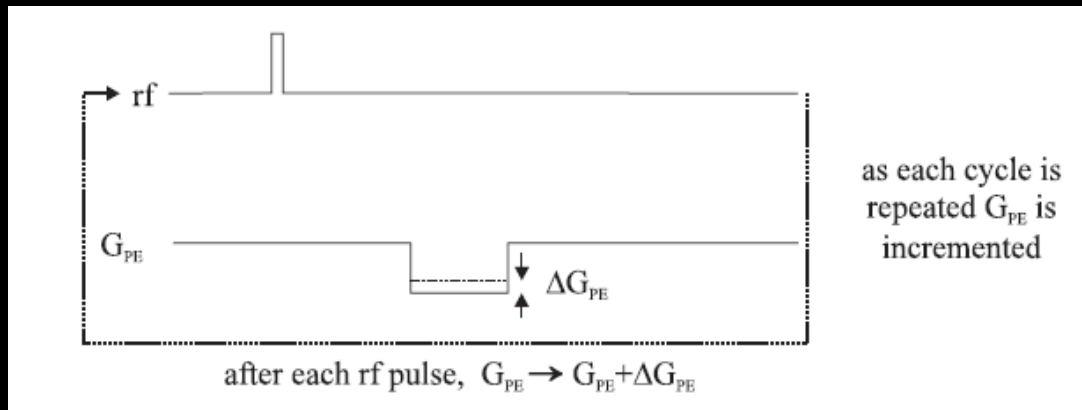
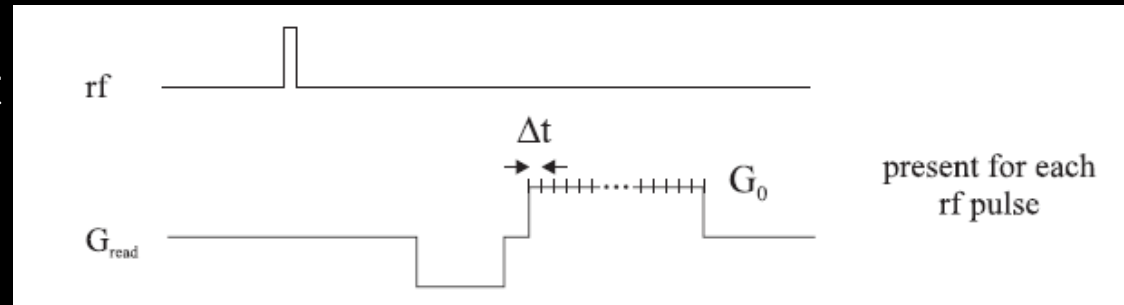


Phase encoding

Frequency encoding

$$k_x = \gamma \cdot G_x \cdot t = \gamma \cdot G_x \cdot \Delta t$$

Encoding phase is accumulated with the progression of time



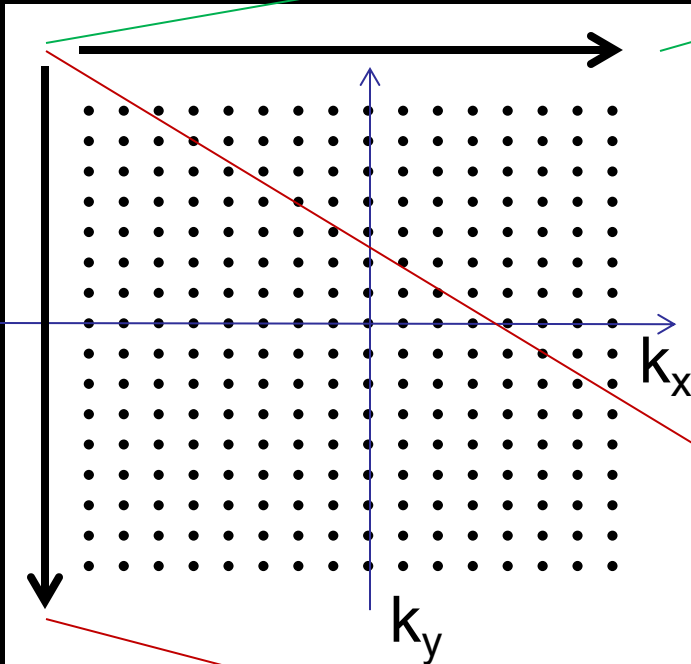
Phase encoding

$$k_y = \gamma \cdot G_y \cdot t$$

$$= \gamma \cdot \Delta G_y \cdot \tau$$

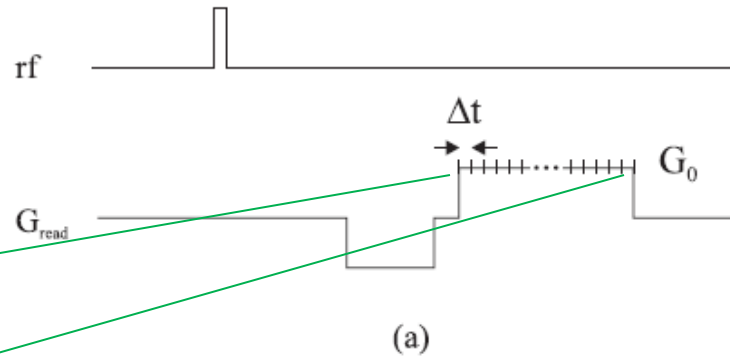
Encoding phase is accumulated in steps of the phase encoding gradient

Phase encoding

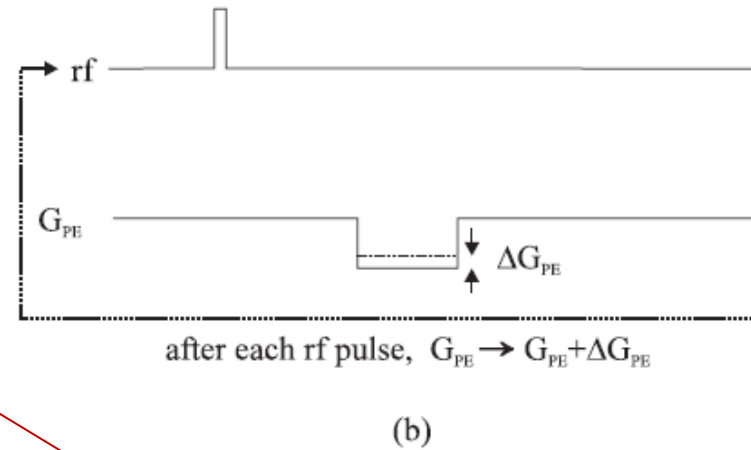


$$k_y = \gamma \cdot G_y \cdot t$$

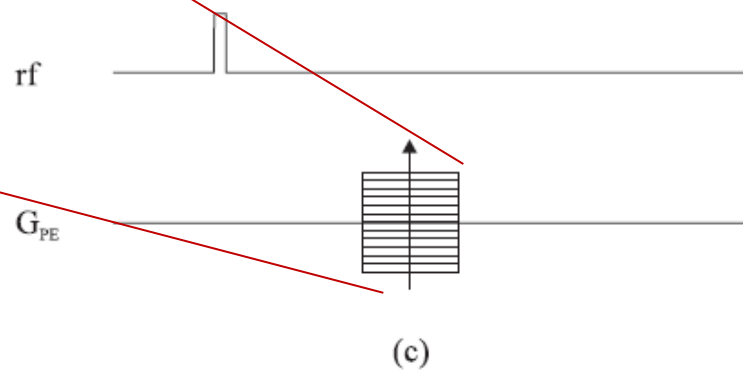
$$= \gamma \cdot \Delta G_y \cdot \tau$$



present for each
rf pulse

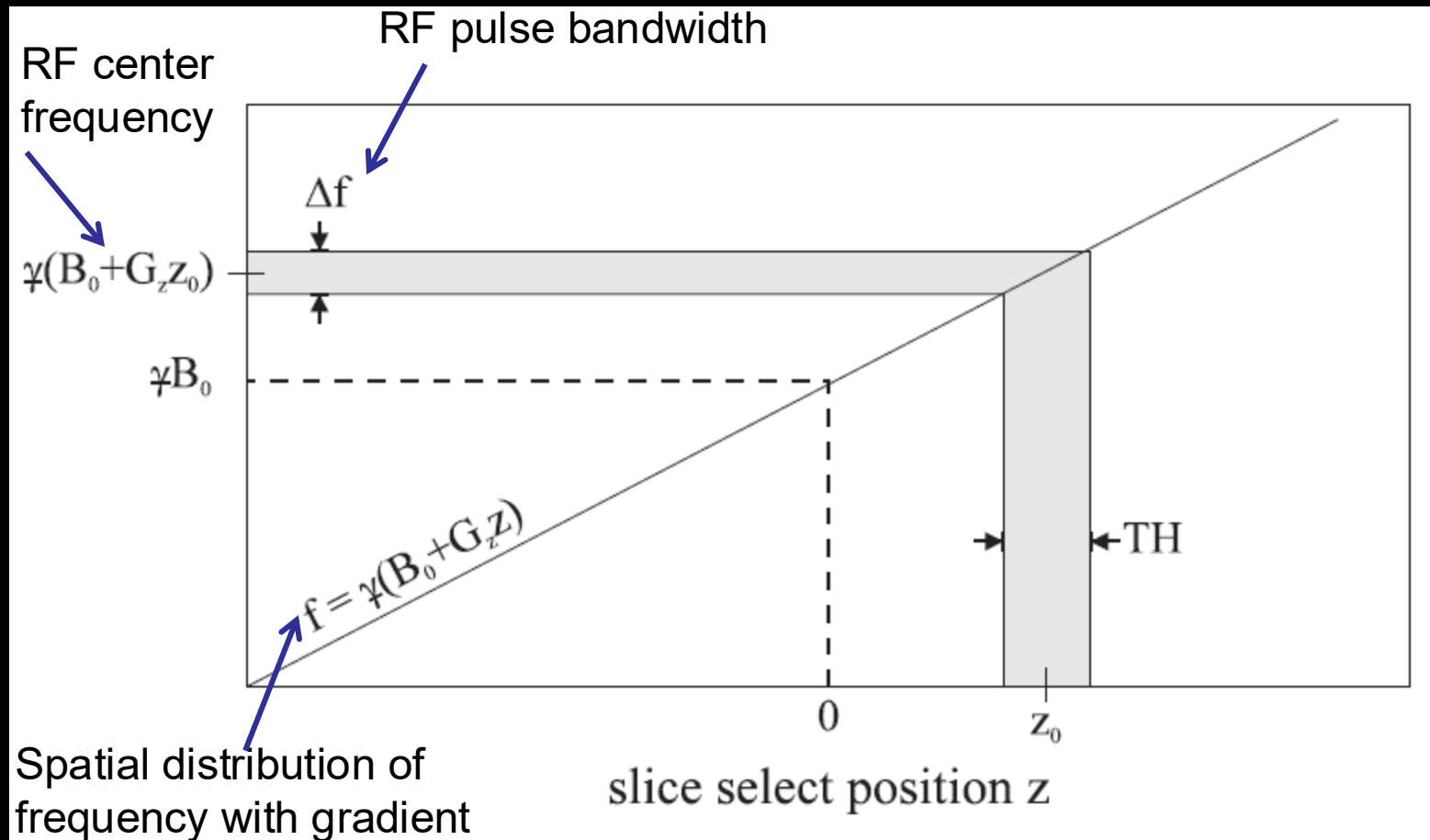
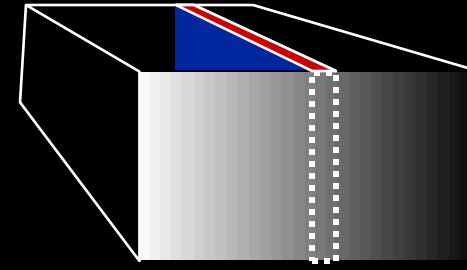


as each cycle is
repeated G_{PE} is
incremented



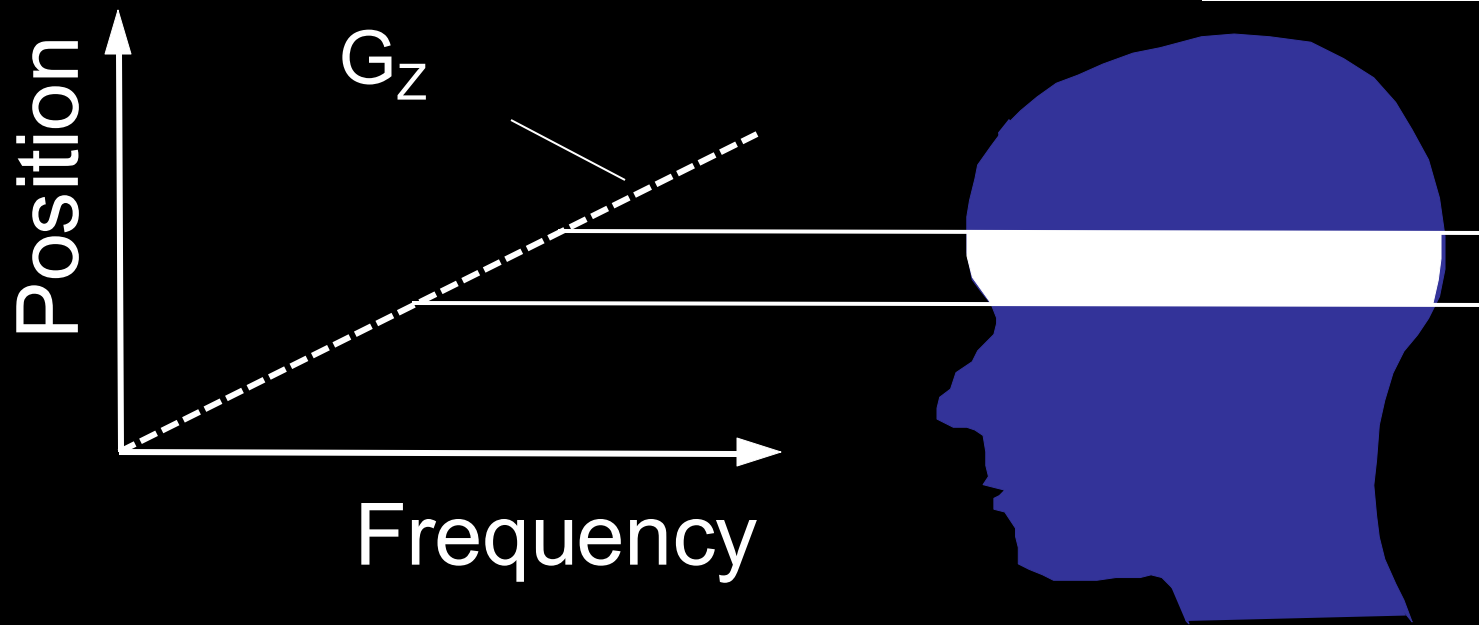
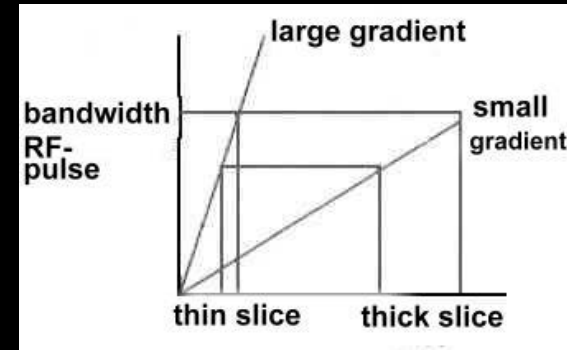
time between
rf pulses is T_R

Slice selection



Note: G_{ss} can be along arbitrary direction

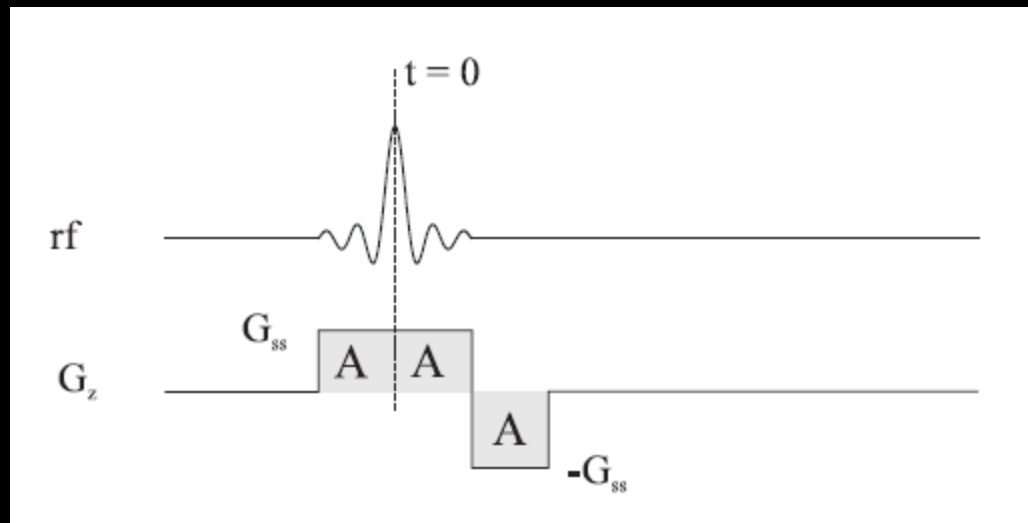
Slice Selection

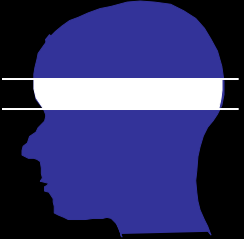


Slice selection gradient refocusing

Spin dephasing during excitation

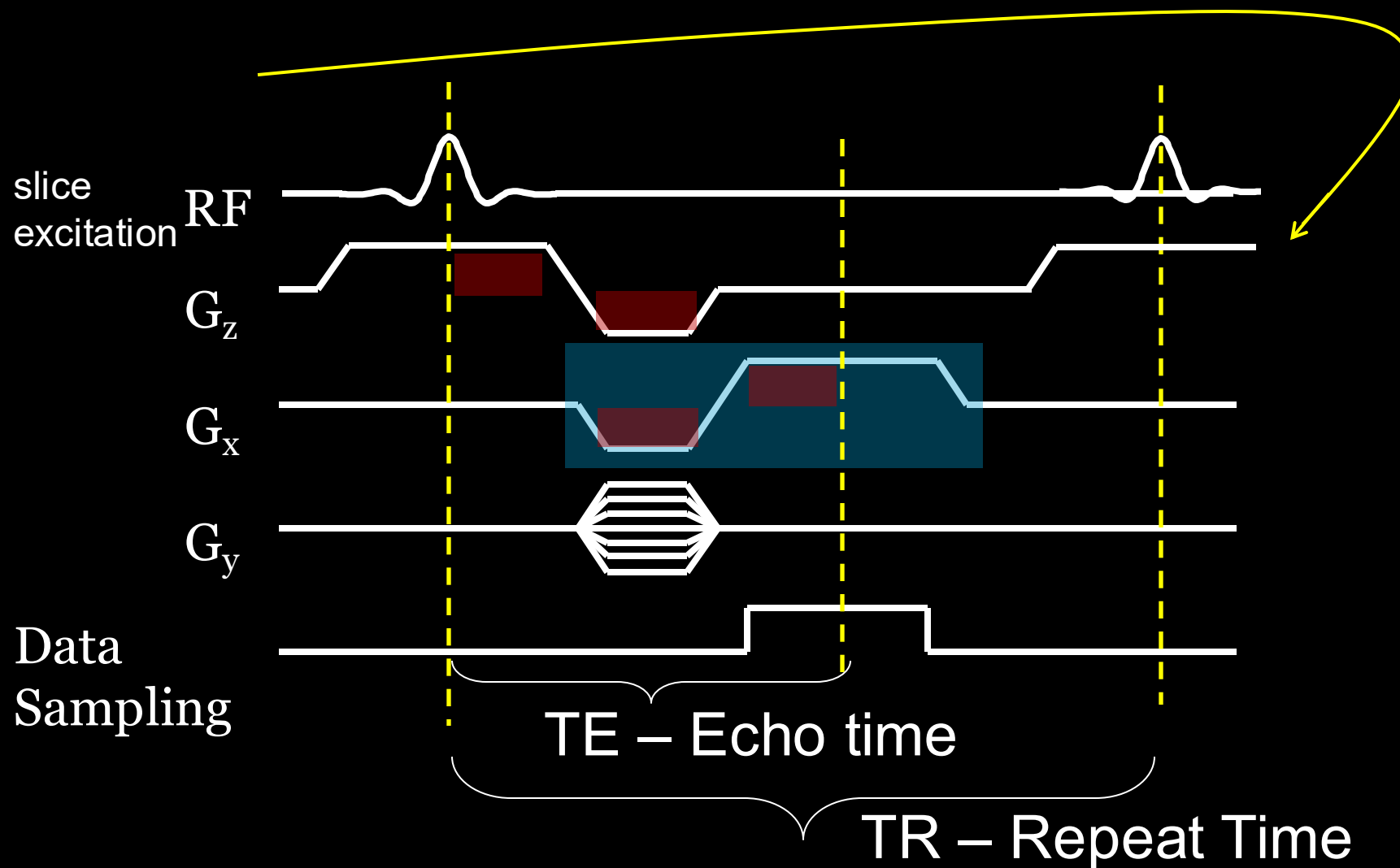
Spins are considered instantaneously tipped at the center of RF pulse.
So, they start getting dephased under the influence of G_{ss} (1/2)





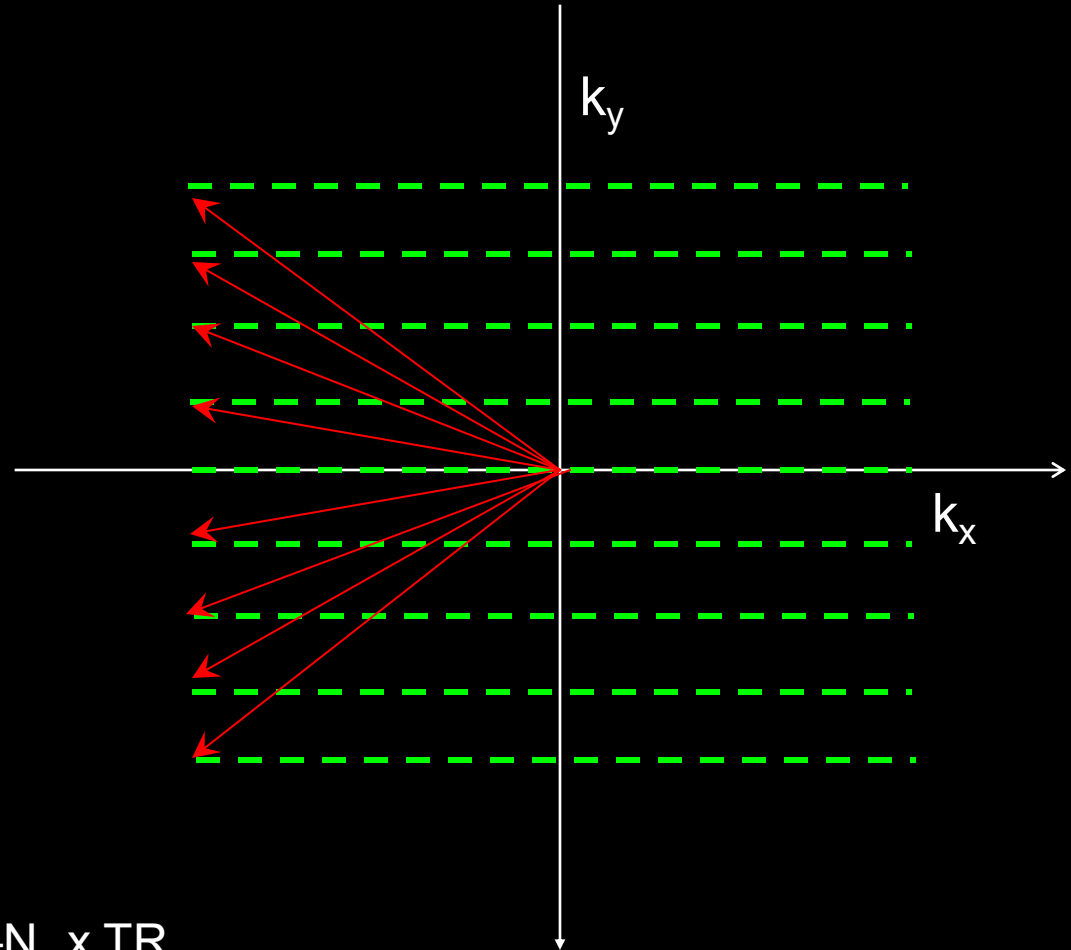
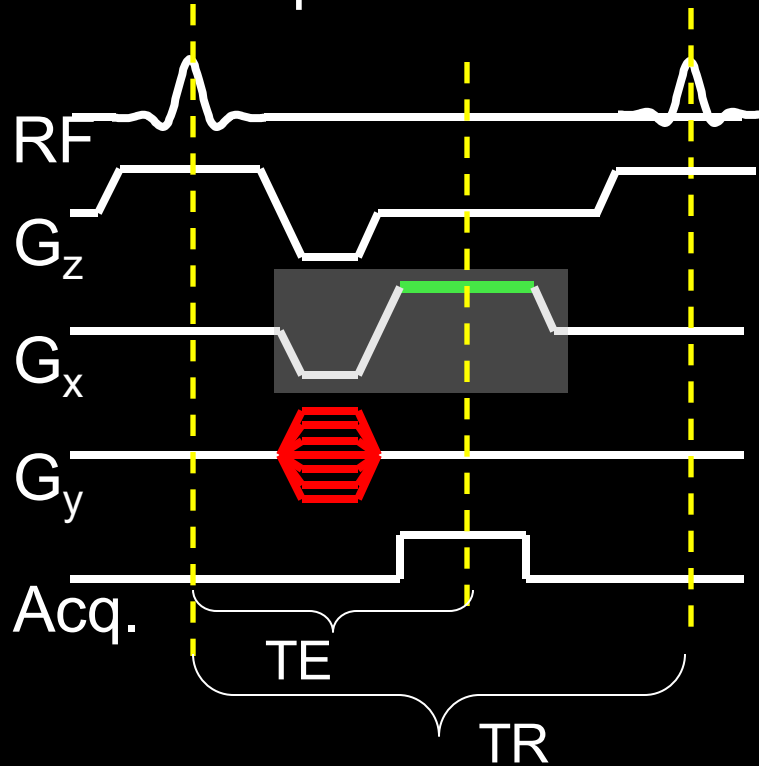
Acquiring the data... line by line

Typical **Gradient Echo** sequence



2D imaging: K-space trajectory

Typical Gradient
Echo sequence

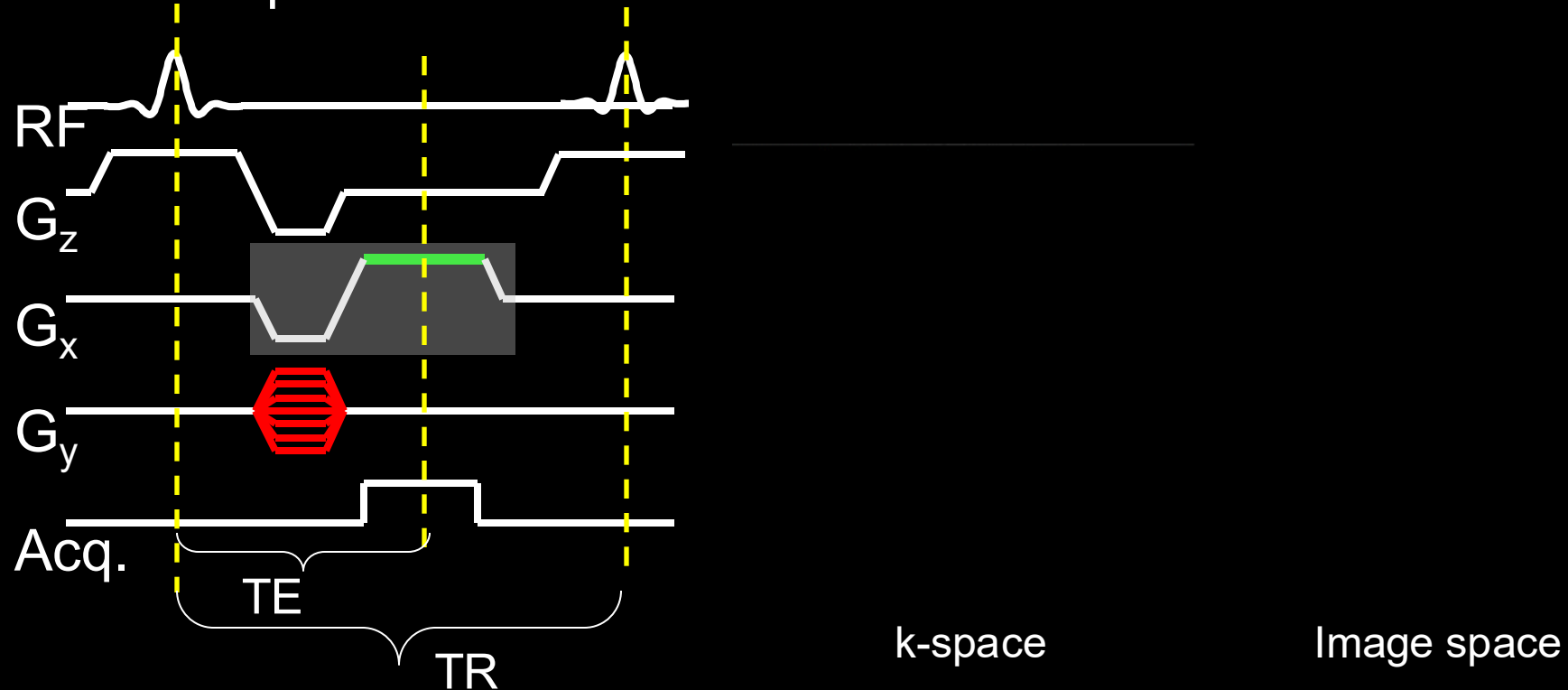


Total imaging time for a single slice $-N_y \times TR$

Sampling of the signal line by line forming the image in
Spatial frequency domain

2D imaging: K-space trajectory

Typical Gradient
Echo sequence

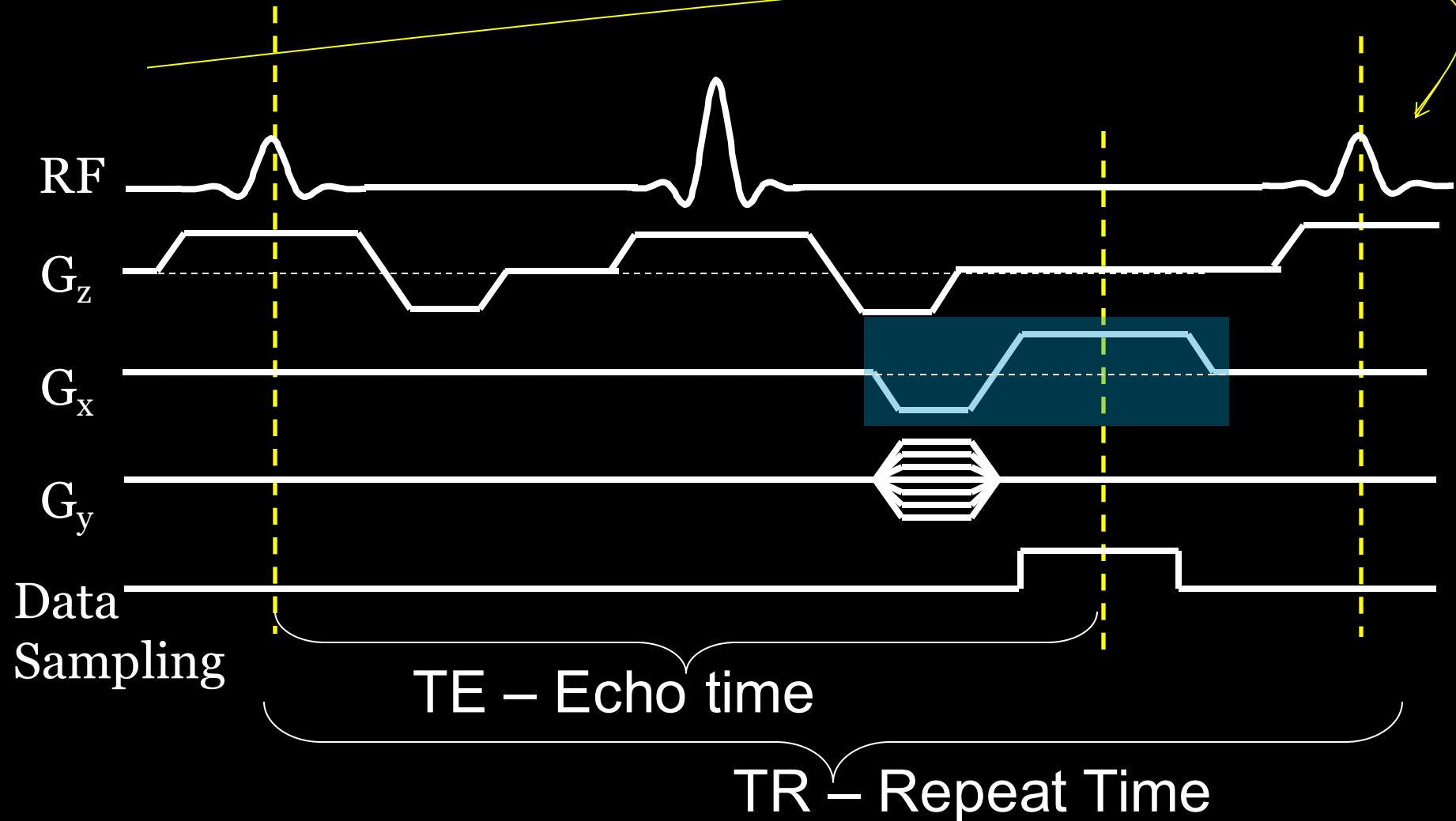


Total imaging time for a single slice $-N_y \times TR$

Sampling of the signal line by line forming the image in
Spatial frequency domain

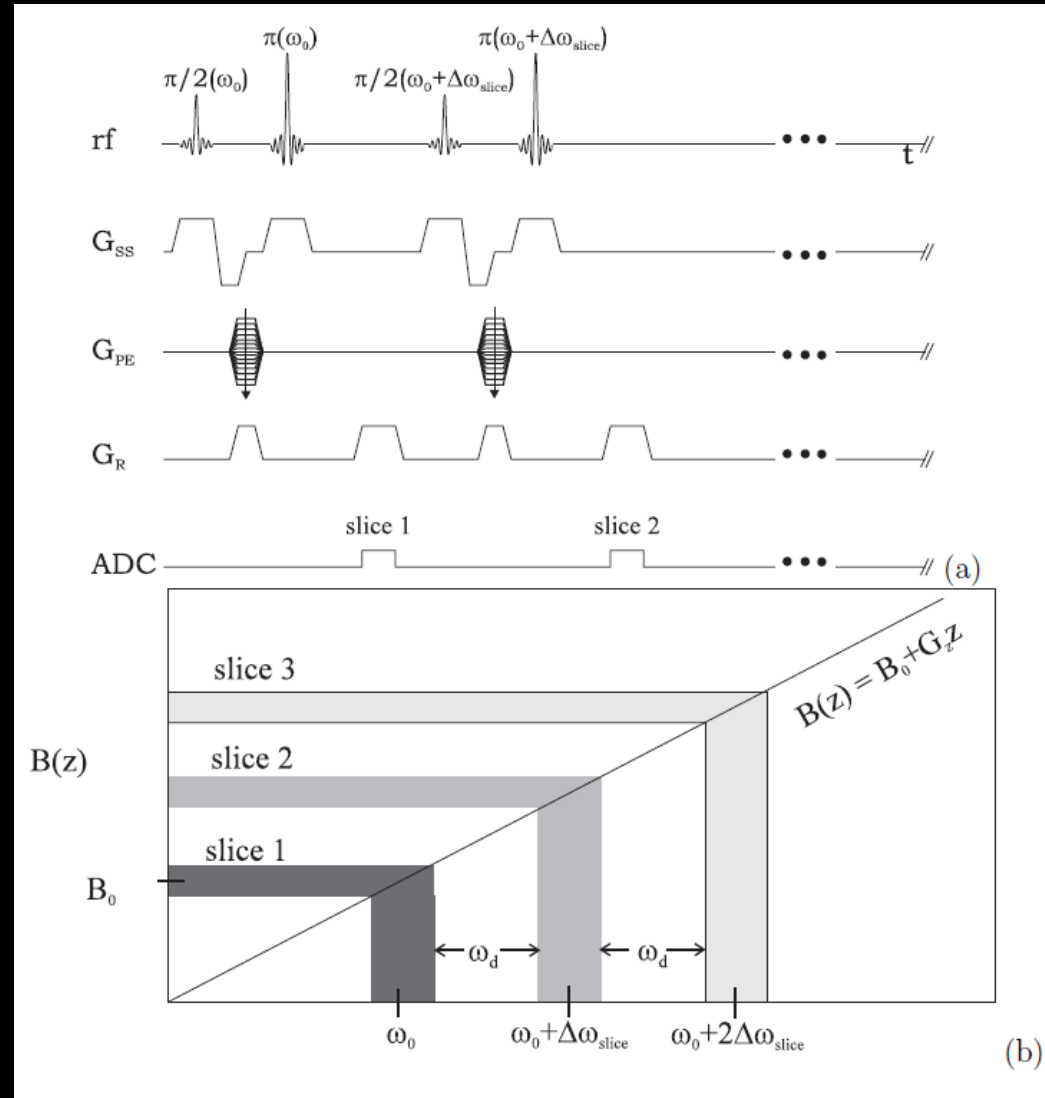
Acquiring the data... line by line

Typical **Spin Echo** sequence

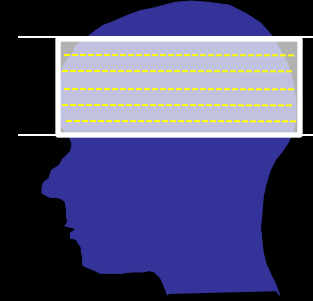


Multi-slice 2D imaging

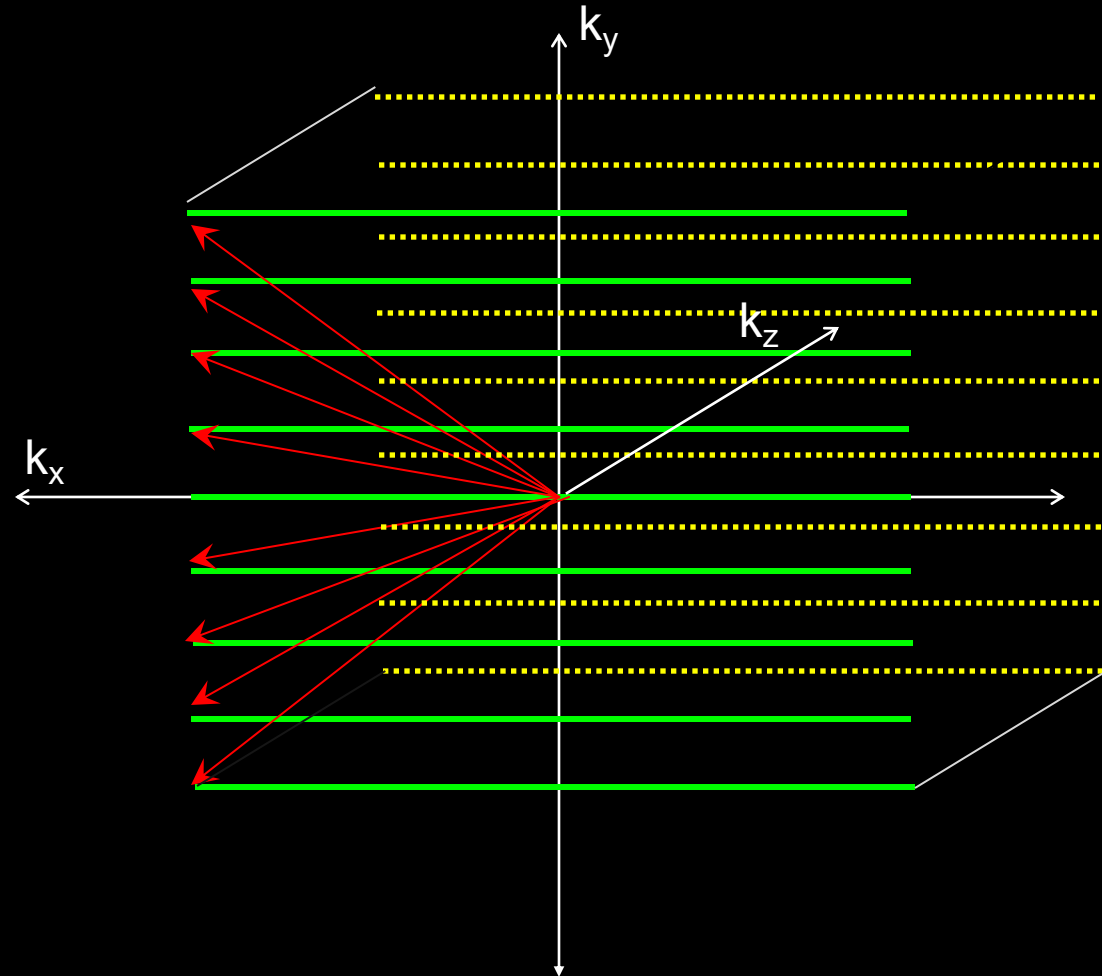
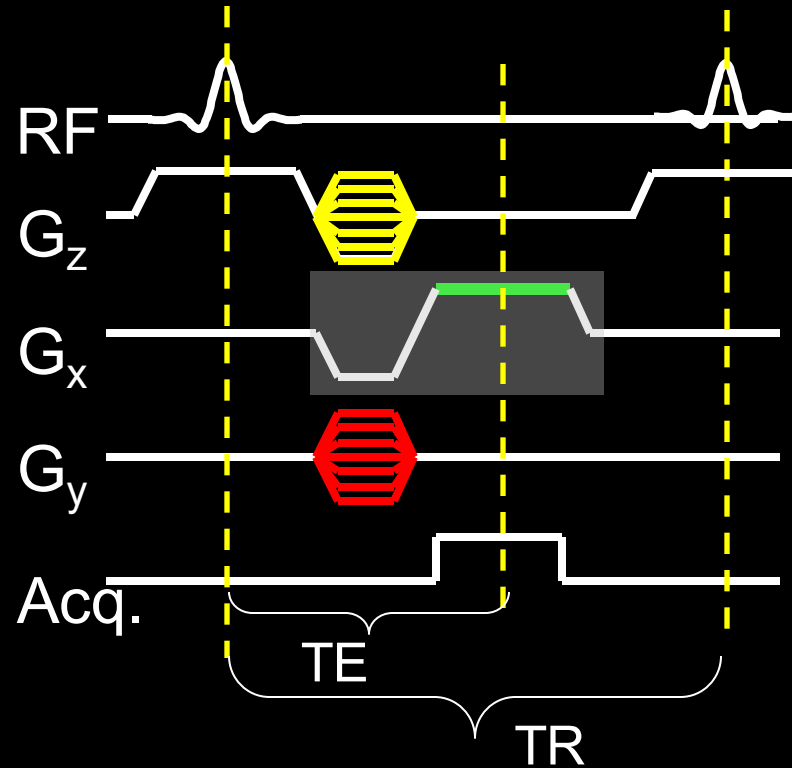
- Same G_{ss} but varying RF center frequency
- Slice gap is usually needed due to imperfect RF frequency profiles
- What is the TR here? (using the dead time)
(Hint: TR can be defined as the interval between Readouts for the same slice)



3D imaging



3D GRE - with slab excitation and partition encoding



Total Imaging time – $N_z \times N_y \times TR$

Slab Excitation and partition encoding in the z direction.

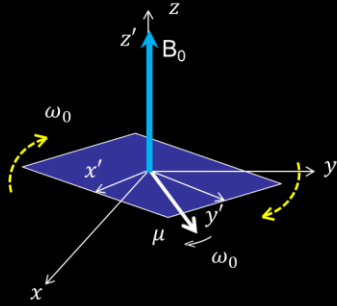
Basic tissue contrast mechanisms in MRI

Basic Contrast Mechanisms: Relaxation effects

The ability to distinguish one tissue from another, i.e., contrast, is a key element in defining an imaging modality's utility

While MRI provides numerous ways of generating contrast between tissues, there are a couple of basic contrast mechanisms that are intrinsic in any magnetic resonance experiment

Basic Contrast Mechanisms: Relaxation effects



Once the magnetization is tipped by the action of an rf pulse, it returns to its equilibrium position eventually

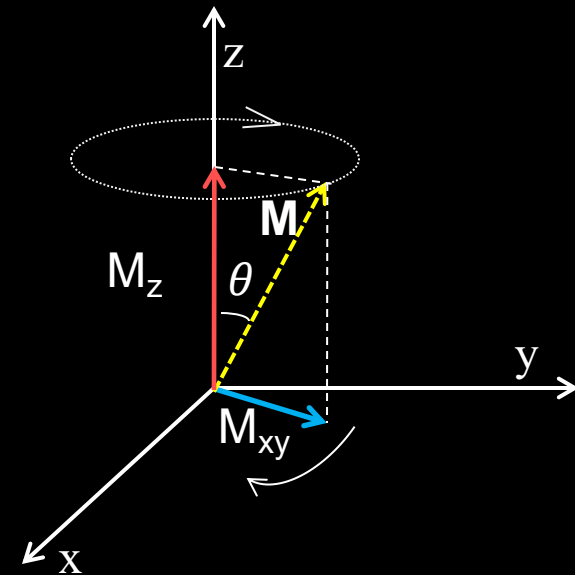
This process of return to equilibrium was empirically determined by Felix Bloch and is given in a familiar form of differential equation

Basic Contrast Mechanisms: Relaxation effects

$$\frac{d\vec{M}}{dt} = \gamma \cdot \vec{M} \times B + \frac{M_0 - M_z}{T_1} \hat{z} - \frac{M_{xy}}{T_2} \hat{\phi}$$

M_0 is the equilibrium magnetization

M_{xy} is the transverse magnetization

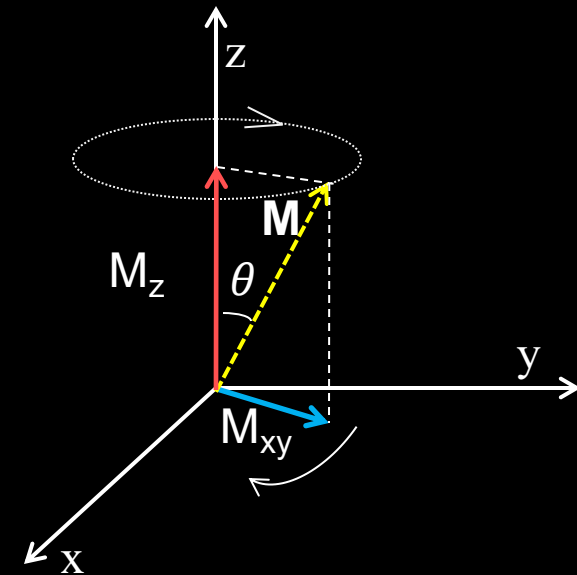
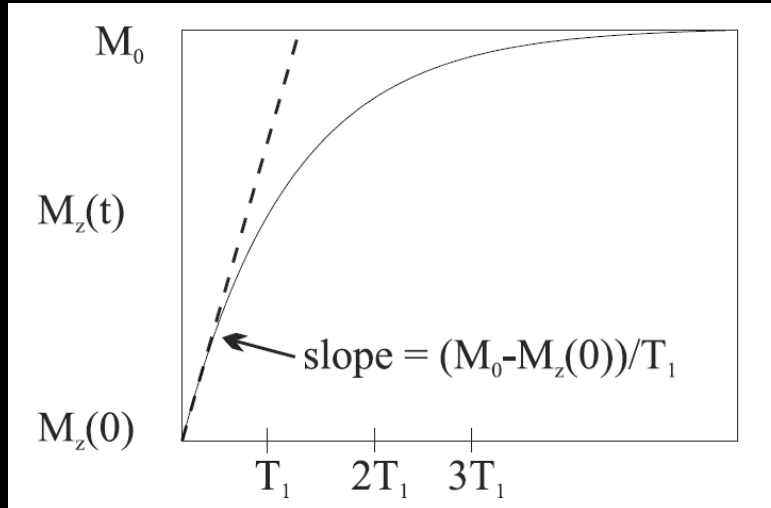


Basic Contrast Mechanisms: Relaxation effects

Longitudinal Magnetization - Recovery

$$\frac{dM_z}{dt} = \frac{M_0 - M_z}{T_1}$$

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$



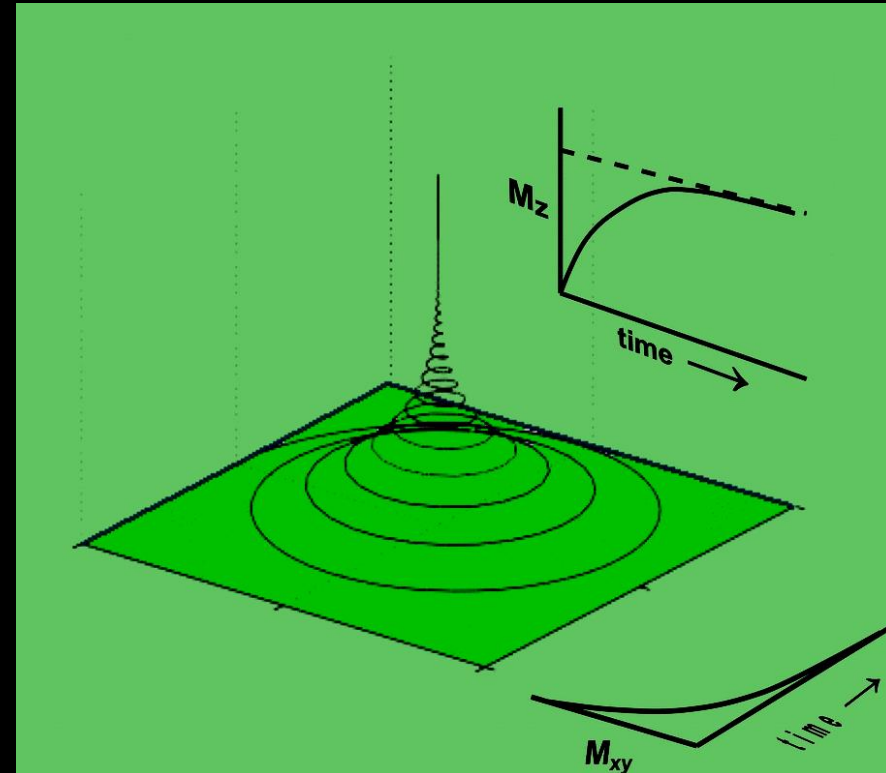
Basic Contrast Mechanisms: Relaxation effects

Longitudinal Magnetization - Recovery

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$

The recovery is characterized by time T_1 , also referred to as spin-lattice relaxation

T_1 is the time it takes the magnetization to recover to 63.2% of its equilibrium value M_0 .



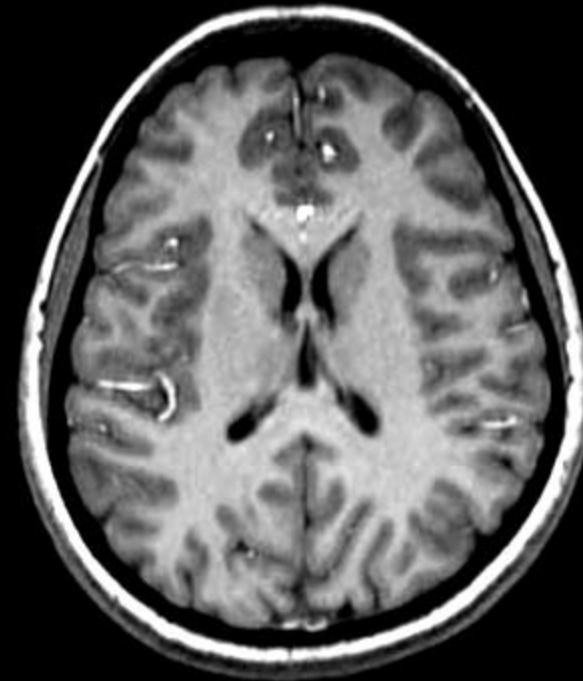
Basic Contrast Mechanisms: Relaxation effects

Longitudinal Magnetization - Recovery

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$

The recovery is characterized by time T_1 , also referred to as spin-lattice relaxation

T_1 is the time it takes the magnetization to recover to 63.2% of its equilibrium value M_0 .



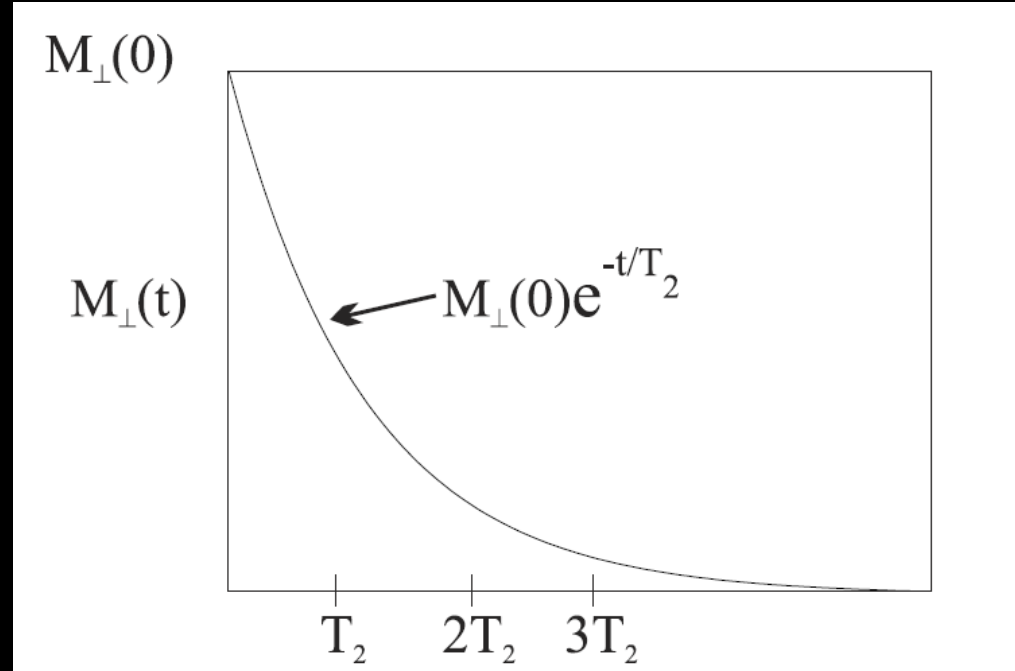
Basic Contrast Mechanisms: Relaxation effects

Transverse Magnetization - Decay

$$\frac{dM_{xy}}{dt} = -\frac{M_{xy}}{T_2} \longrightarrow M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$

The decay time is characterized by the parameter T_2 , also referred to as spin-spin relaxation

T_2 is the time it takes the transverse magnetization to decay to 36.7% of its initial value M_{xy} .



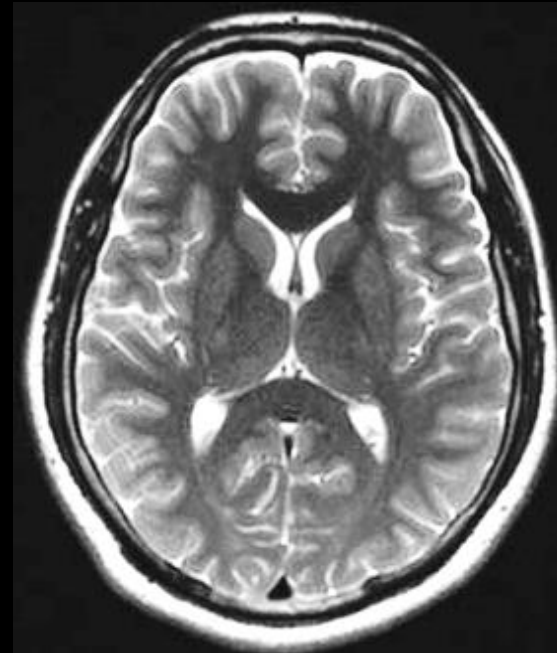
Basic Contrast Mechanisms: Relaxation effects

Transverse Magnetization - Decay

$$\frac{dM_{xy}}{dt} = -\frac{M_{xy}}{T_2} \longrightarrow M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$

The decay time is characterized by the parameter T_2 , also referred to as spin-spin relaxation

T_2 is the time it takes the transverse magnetization to decay to 63.2% of its initial value M_{xy} .



Basic Contrast Mechanisms: Relaxation effects

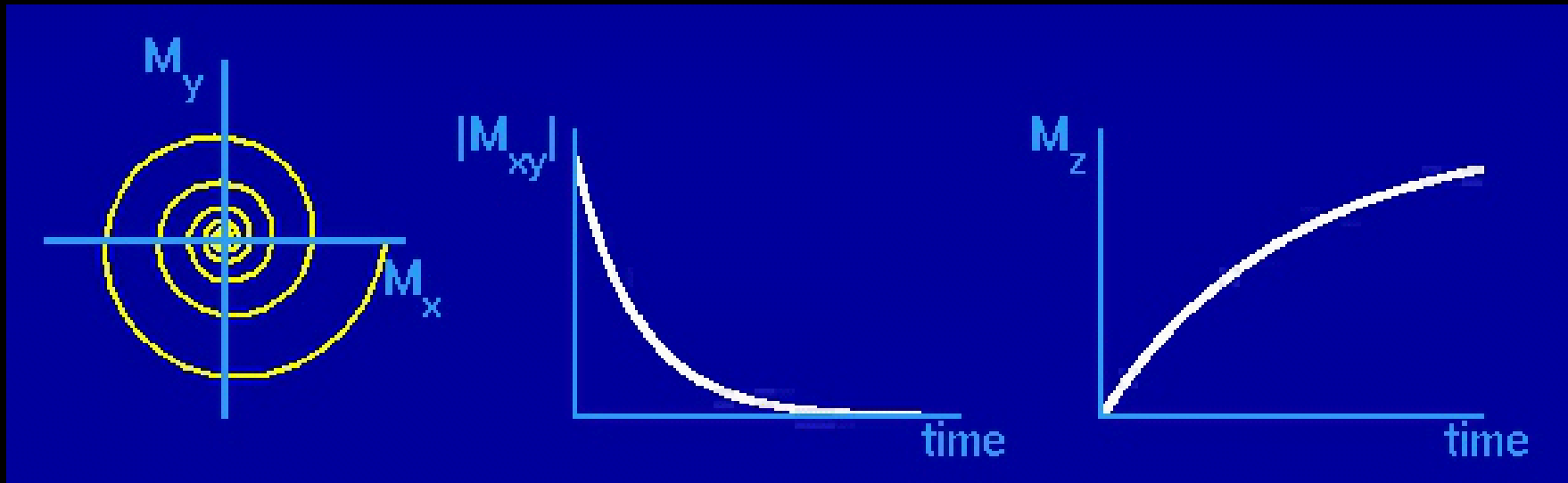
Transverse
Magnetization
Decay

$$M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$

Longitudinal
Magnetization
Recovery

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$

$$T_1 > T_2$$



Basic Contrast Mechanisms: Relaxation effects

Transverse
Magnetization
Decay

$$M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$

Longitudinal
Magnetization
Recovery

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$

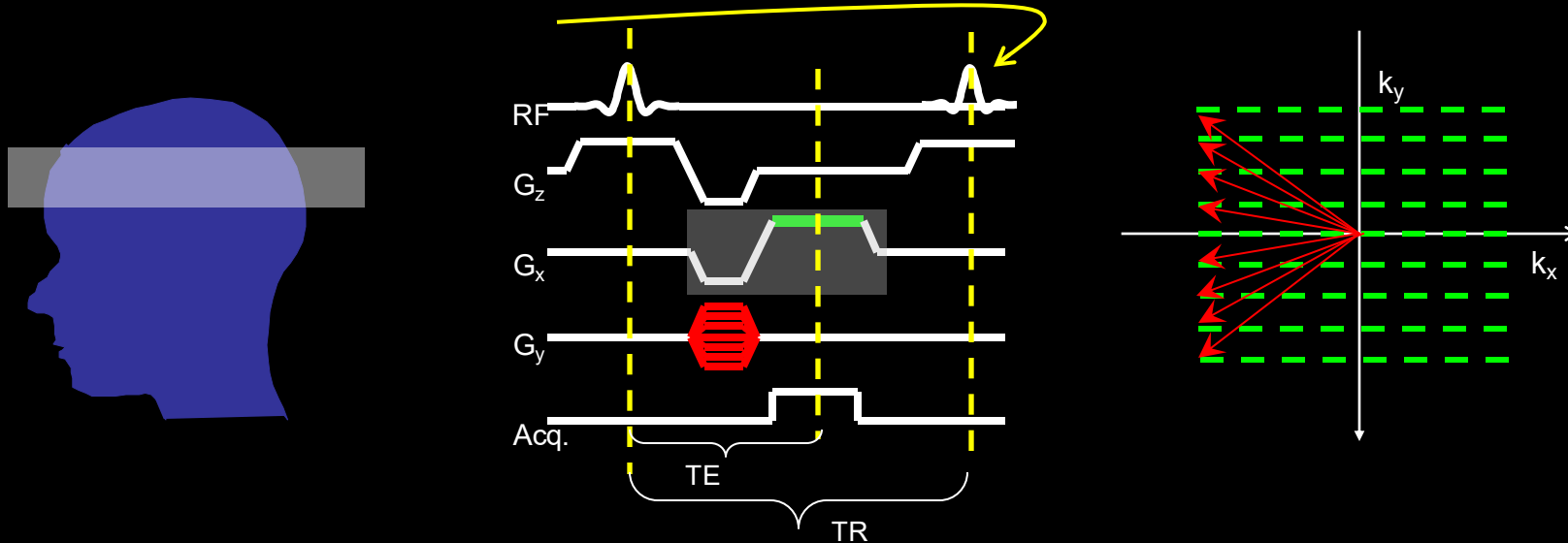
$$T_1 > T_2$$

Tissue	T_1 (ms)	T_2 (ms)
gray matter (GM)	950	100
white matter (WM)	600	80
muscle	900	50
cerebrospinal fluid (CSF)	4500	2200
fat	250	60
blood ³	1200	100-200 ⁴

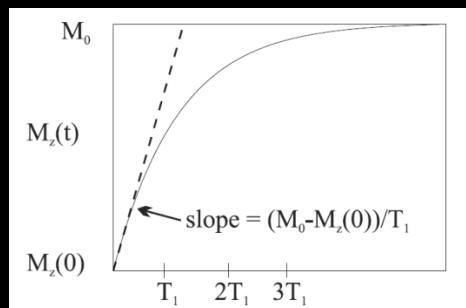
Contrast

Tissue	T_1 (ms)	T_2 (ms)
gray matter (GM)	950	100
white matter (WM)	600	80
muscle	900	50
cerebrospinal fluid (CSF)	4500	2200
fat	250	60
blood ³	1200	100-200 ⁴

Contrast

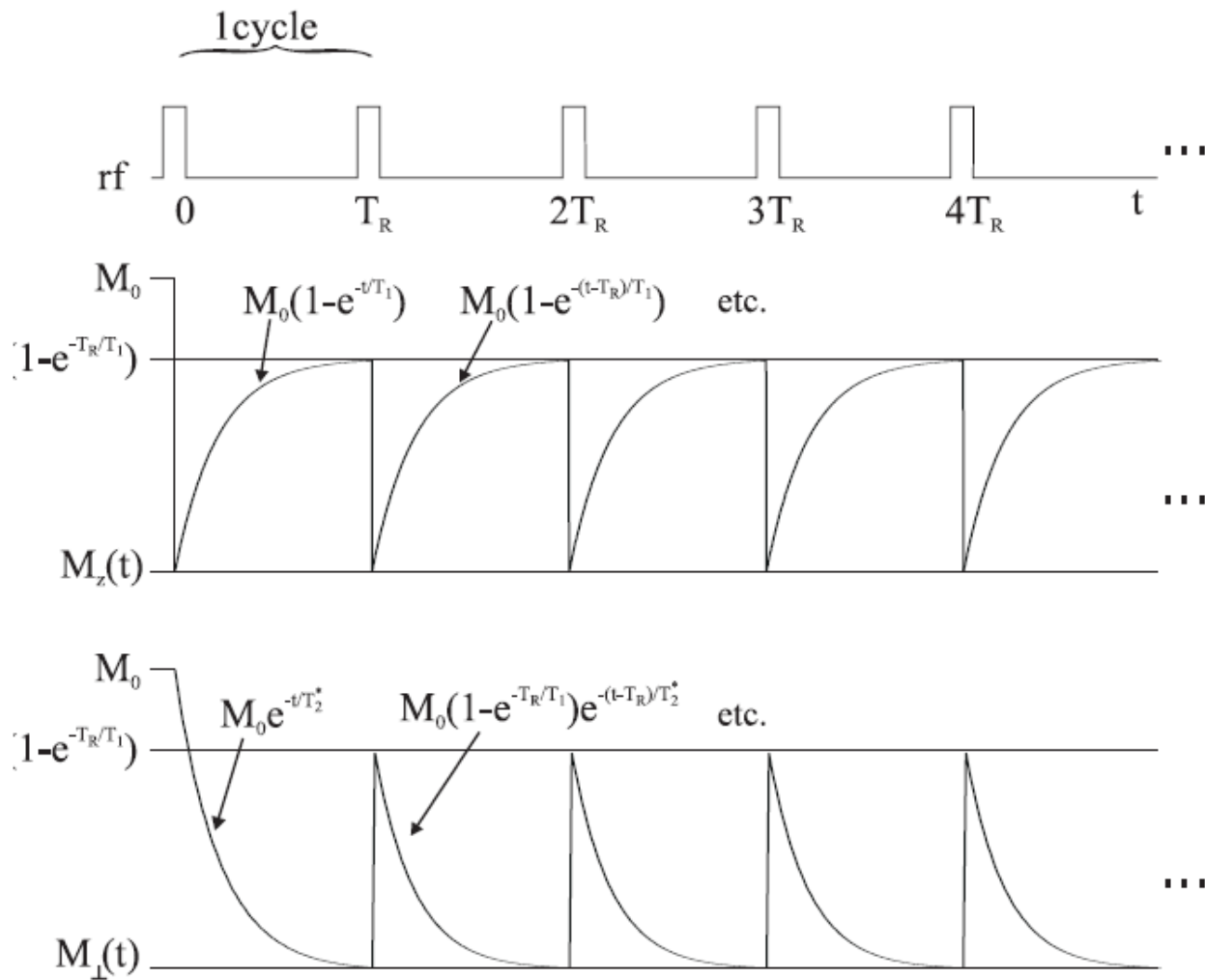


Once you tip the magnetization of a tissue, it takes a finite amount of time for it to recover

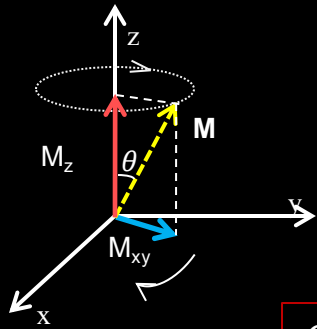


With repeated rf pulse excitations during image acquisition, the time intervals of repetition and acquisition (TR, TE), along with T_1 and T_2 values of different tissue determine the final contrast

Magnetization equations



Magnetization equations



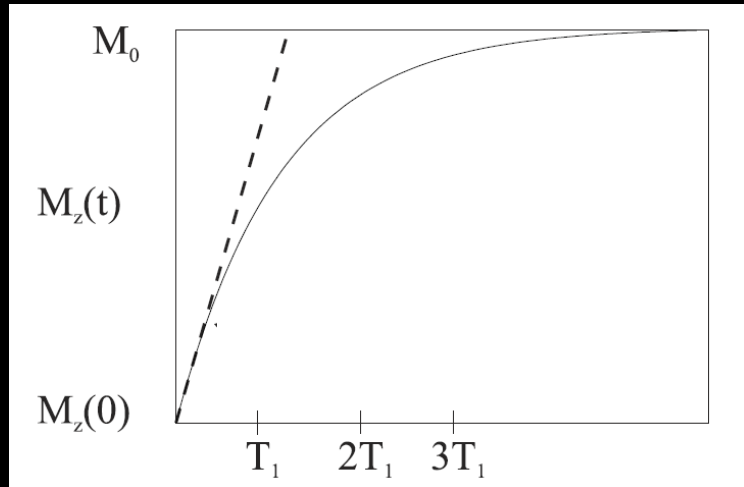
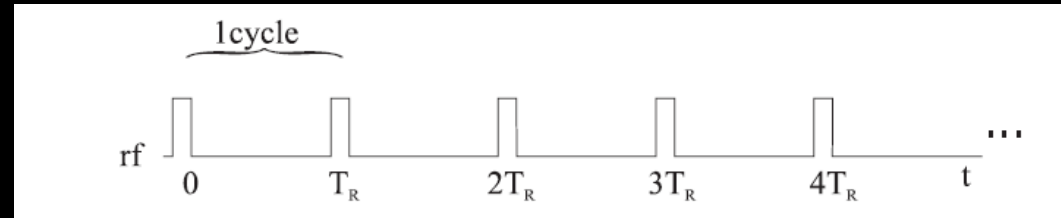
$$\theta = 90^\circ$$

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$

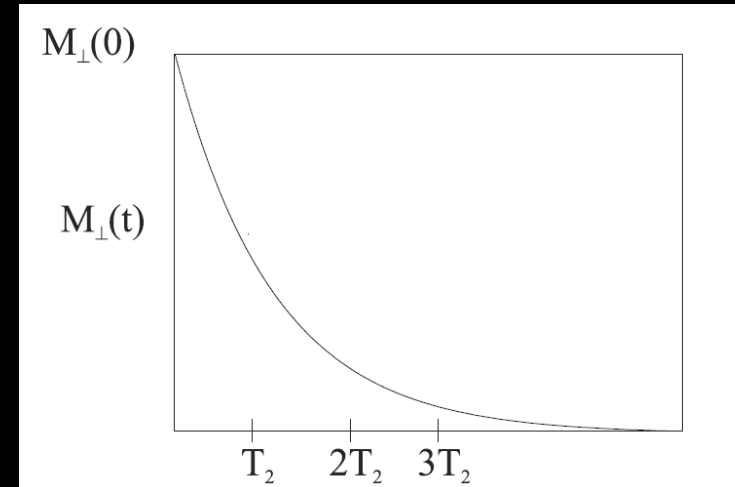
$$M_z(0) = M_0 \cdot \cos(\theta)$$

$$M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$

$$M_{xy}(0) = M_0 \cdot \sin(\theta)$$



$$M_z(TR^-) = M_0 \cdot \left(1 - e^{-\frac{TR}{T_1}}\right)$$

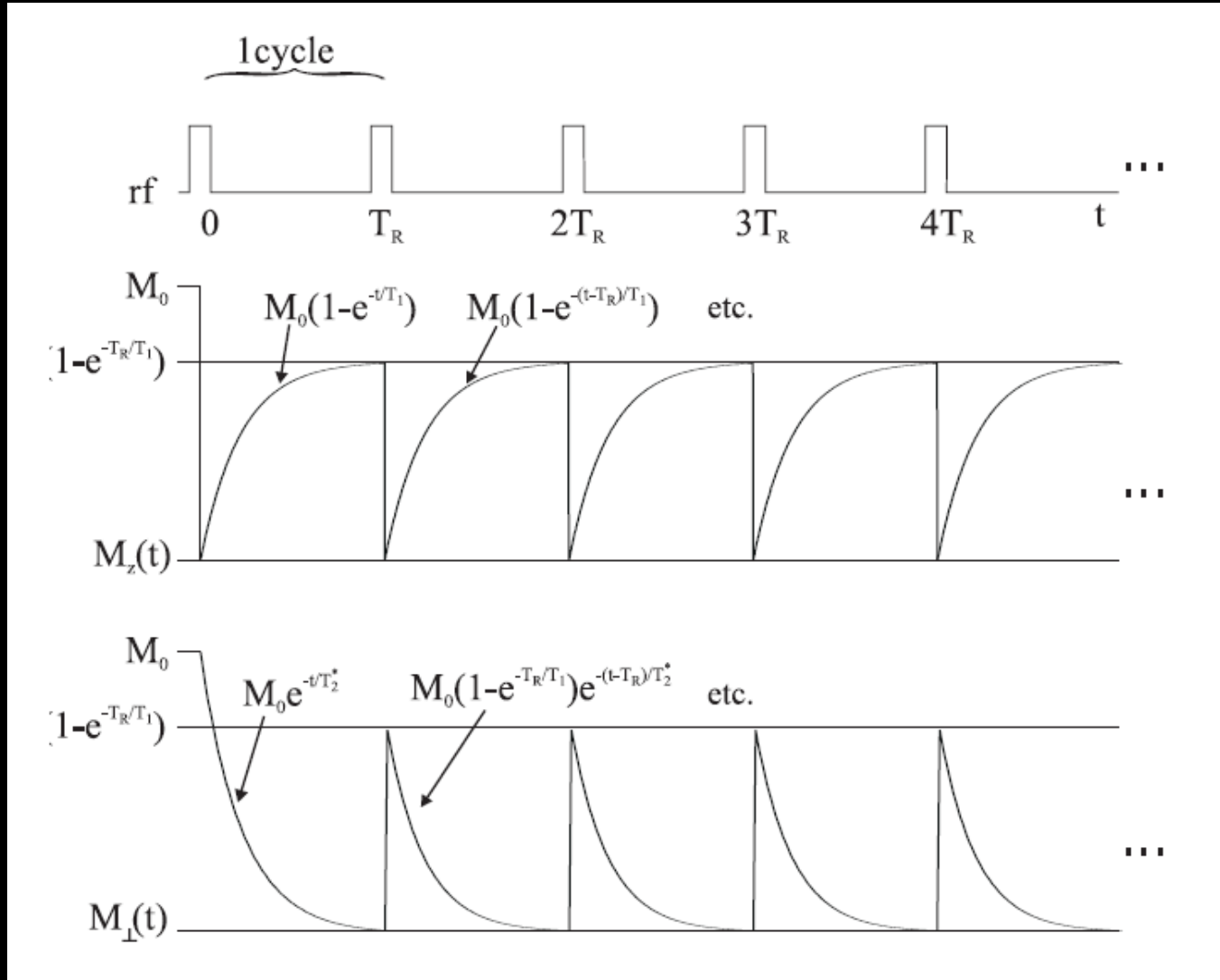


$$M_{xy}(TE) = M_0 \cdot \left(1 - e^{-\frac{TR}{T_1}}\right) \cdot e^{-\frac{TE}{T_2}}$$

$$M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$

$$TR \gg T_2$$

Magnetization equations

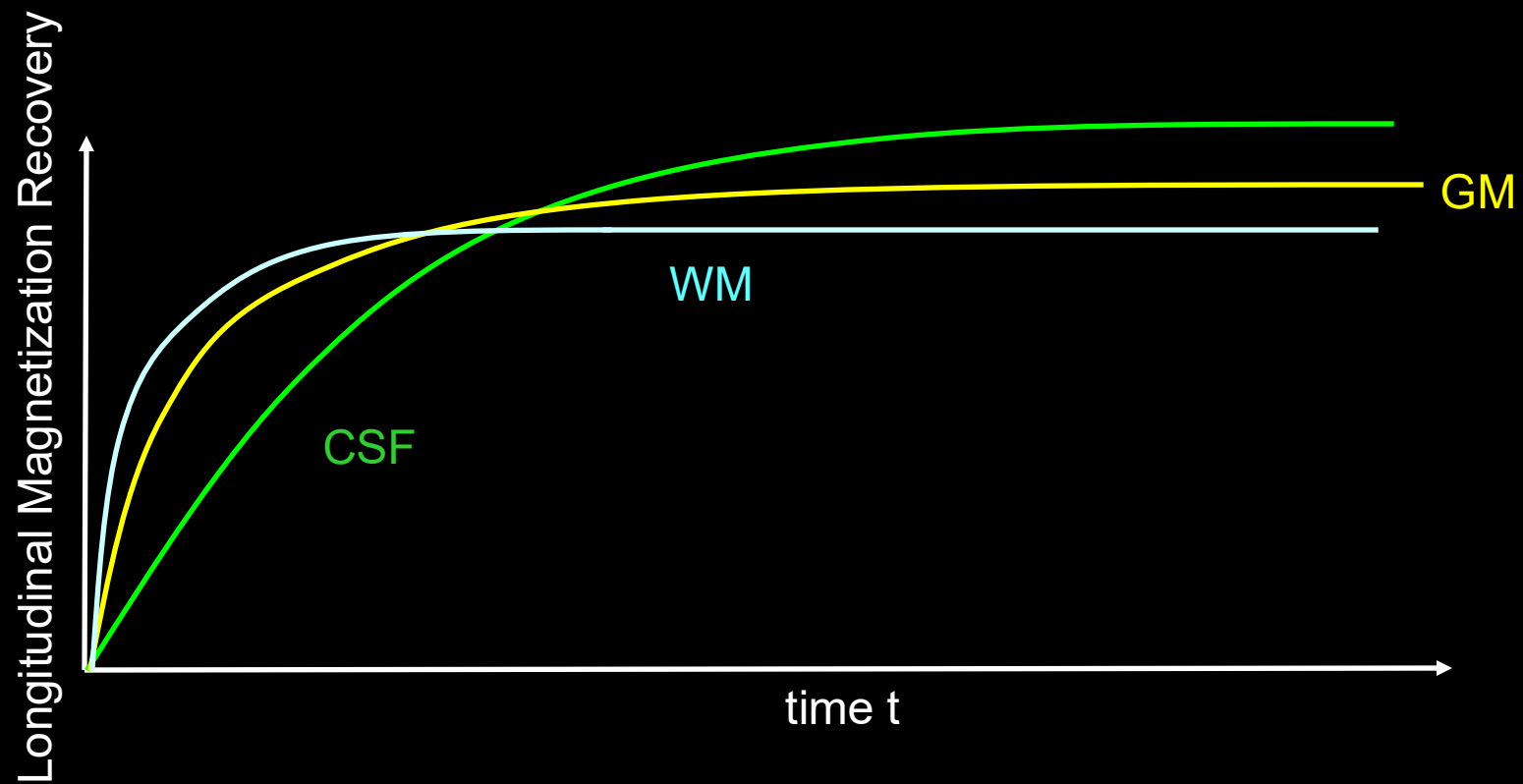


Take-aways from
the math

Steady state
reached by the
second rf pulse

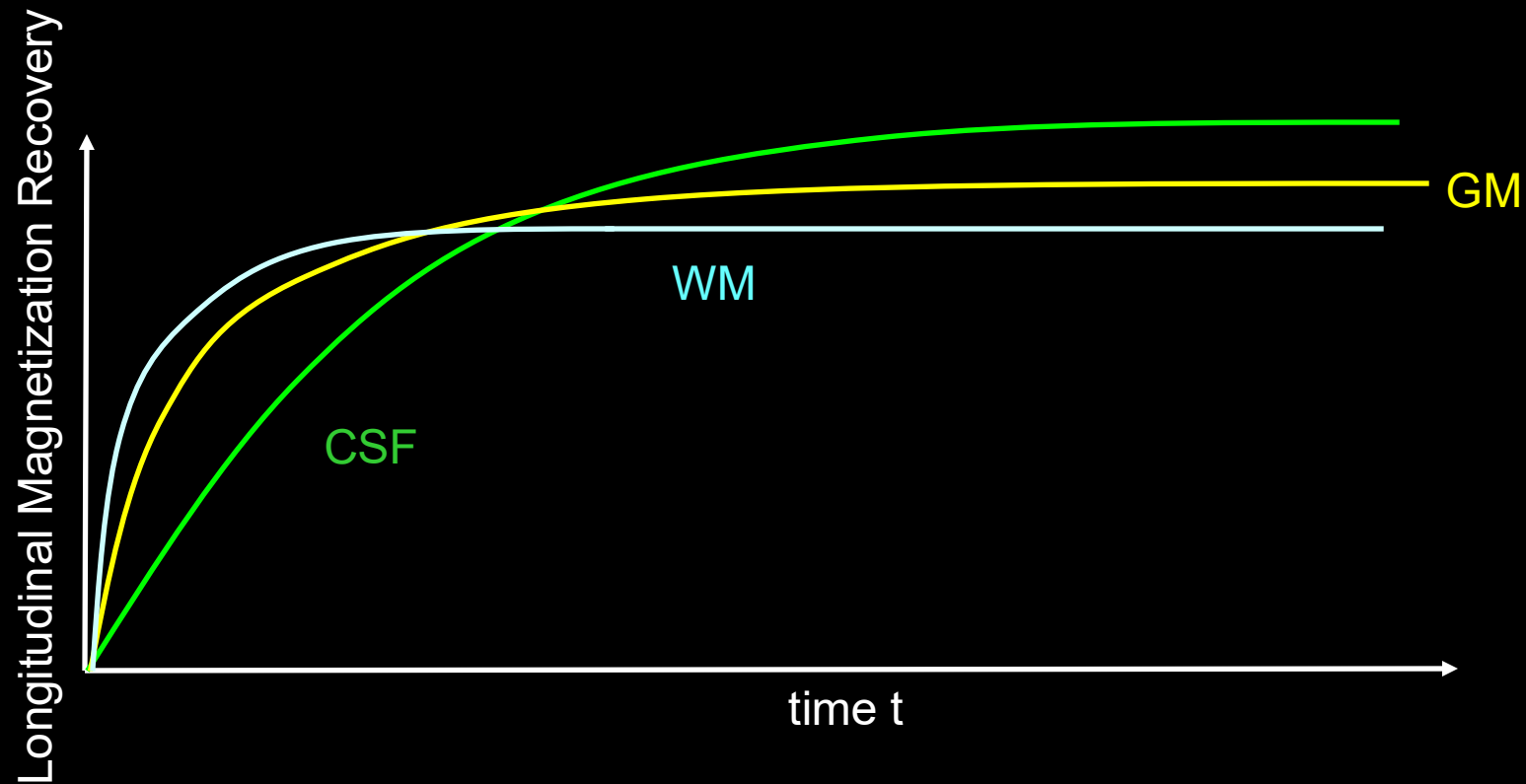
Steady state
meaning... $M_z(TR)$
and $M_{xy}(TE)$ are
constant

Contrast

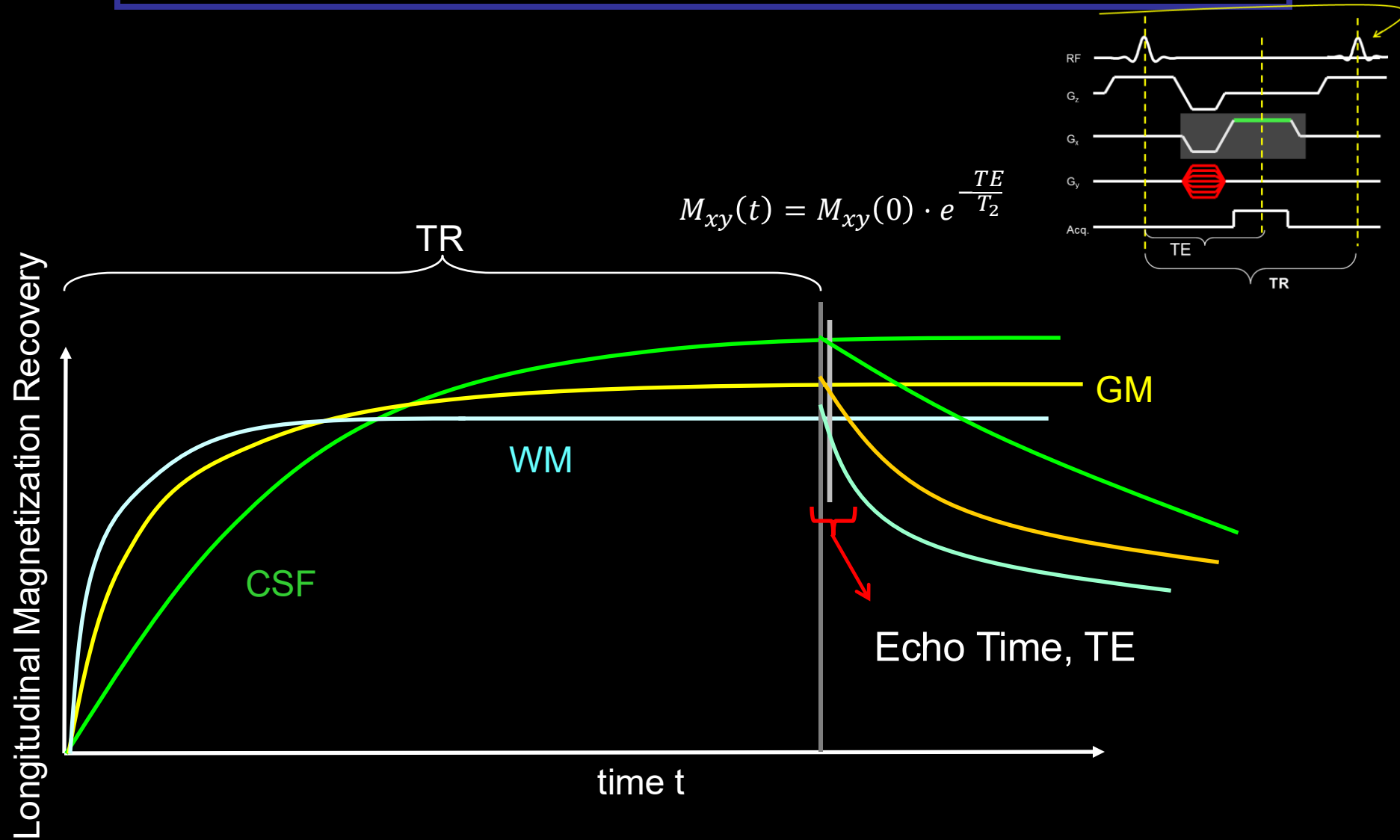


Contrast

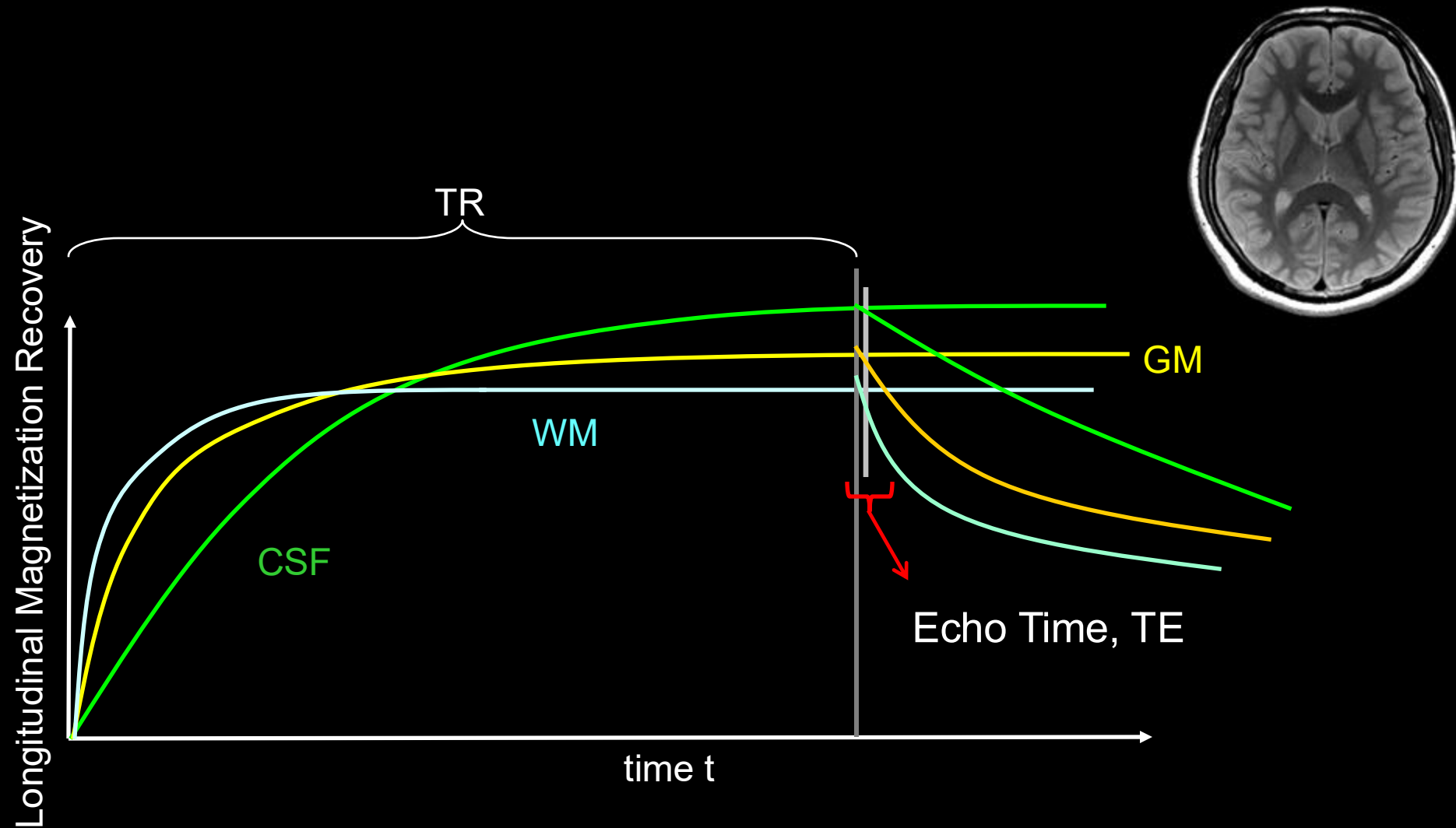
Different tissues have different amounts of water and hence different densities of ^1H protons within them.



Contrast

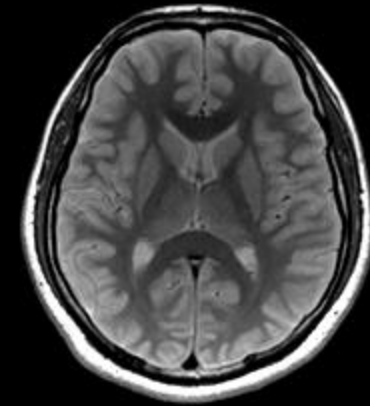
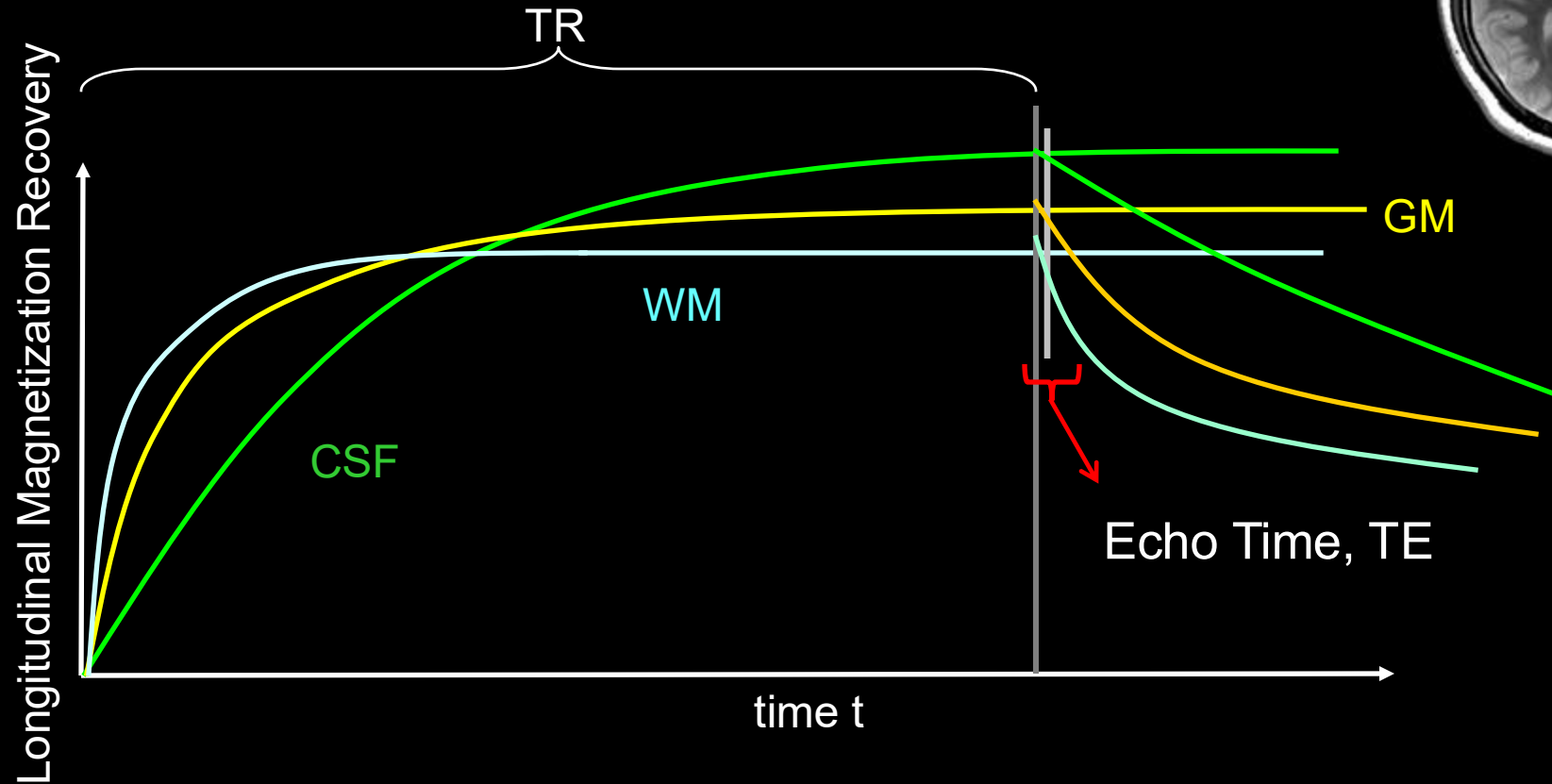


Contrast... ρ_0 weighting

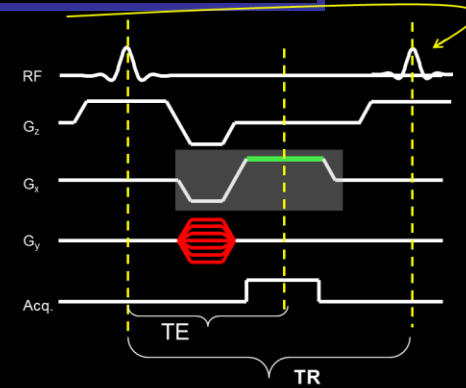
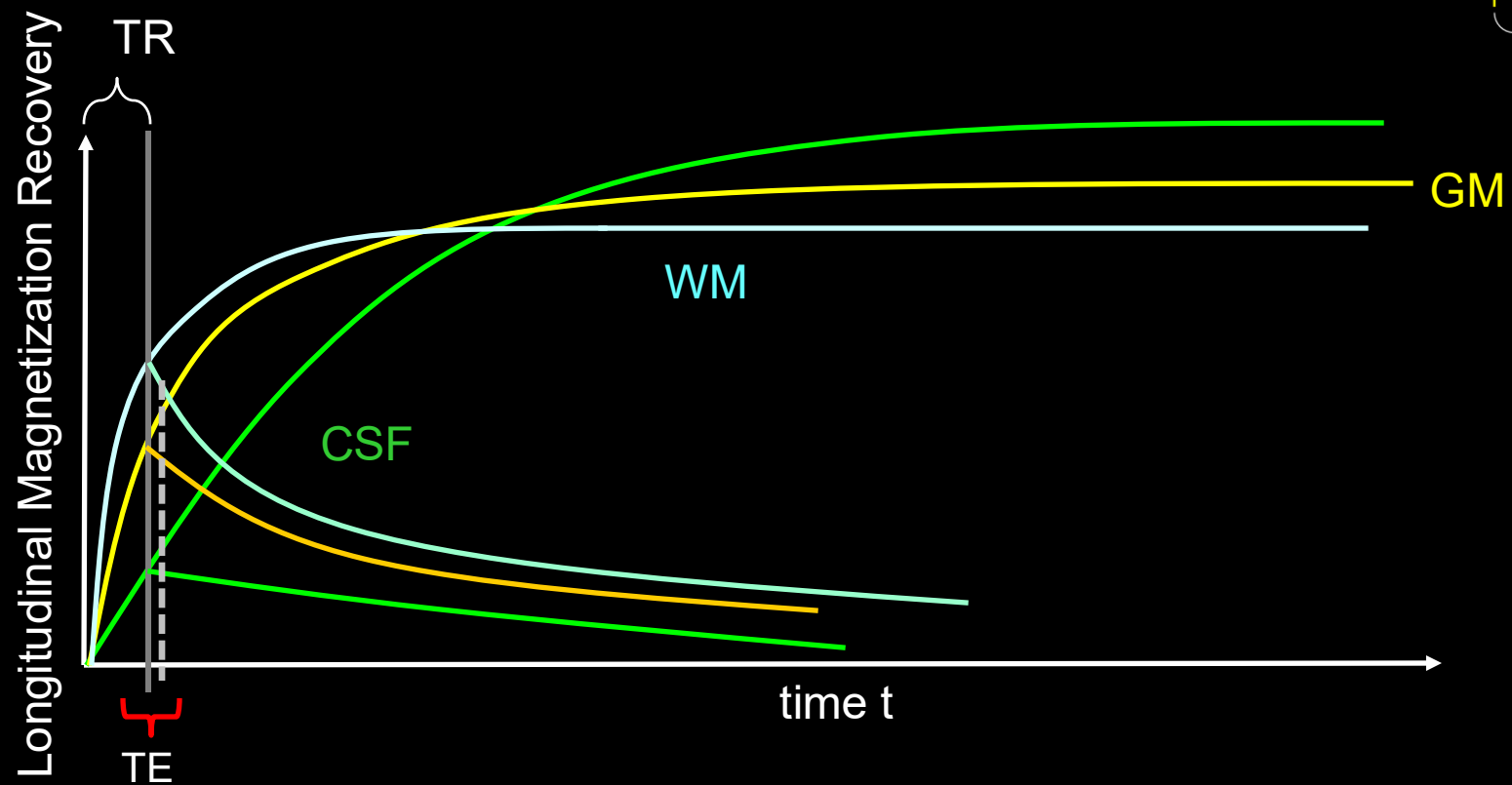


Contrast... ρ_0 weighting

Long repetition time -TR
Short echo time - TE

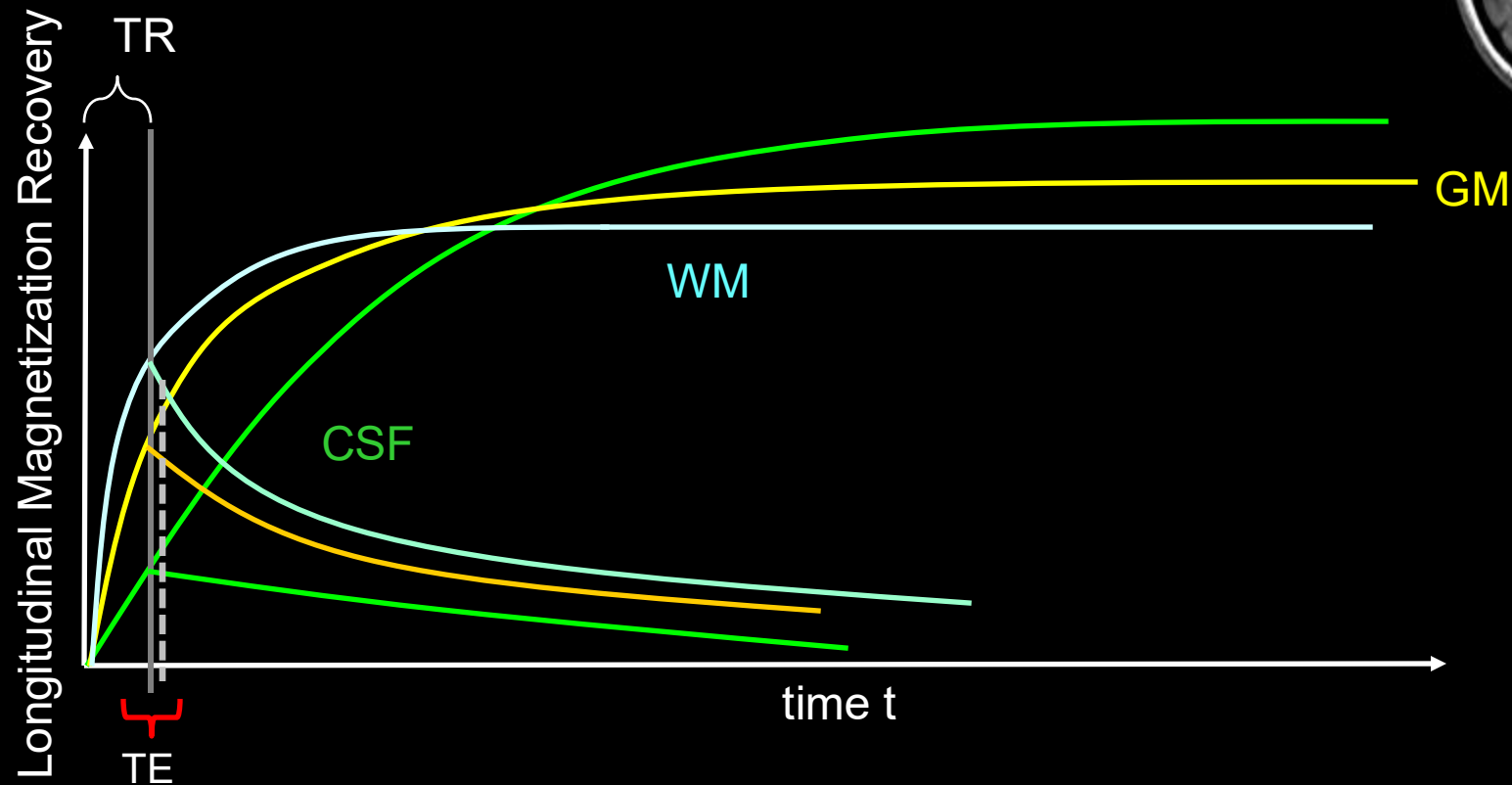
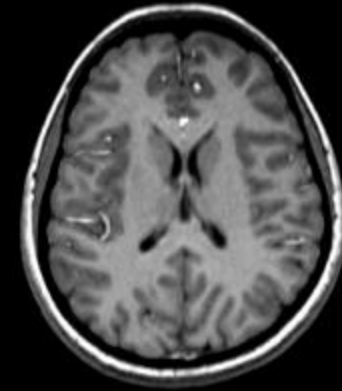


Contrast

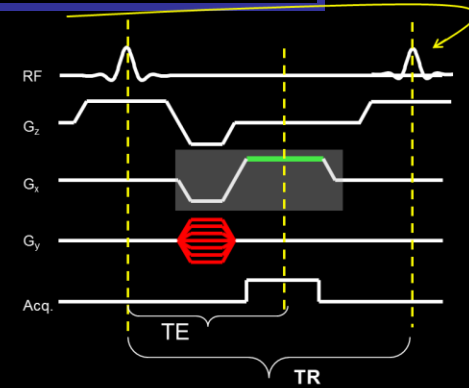
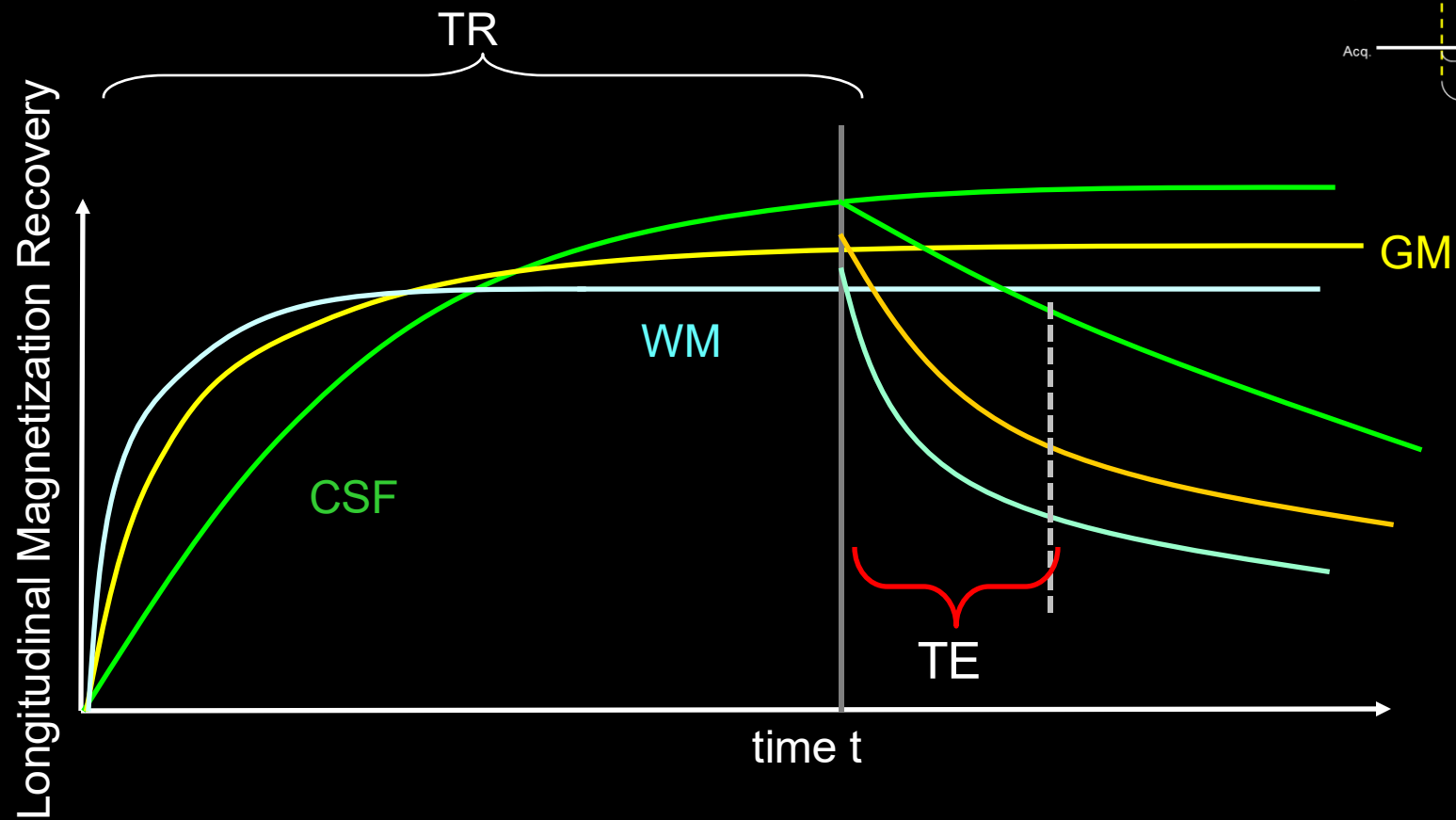


Contrast... T_1 weighting

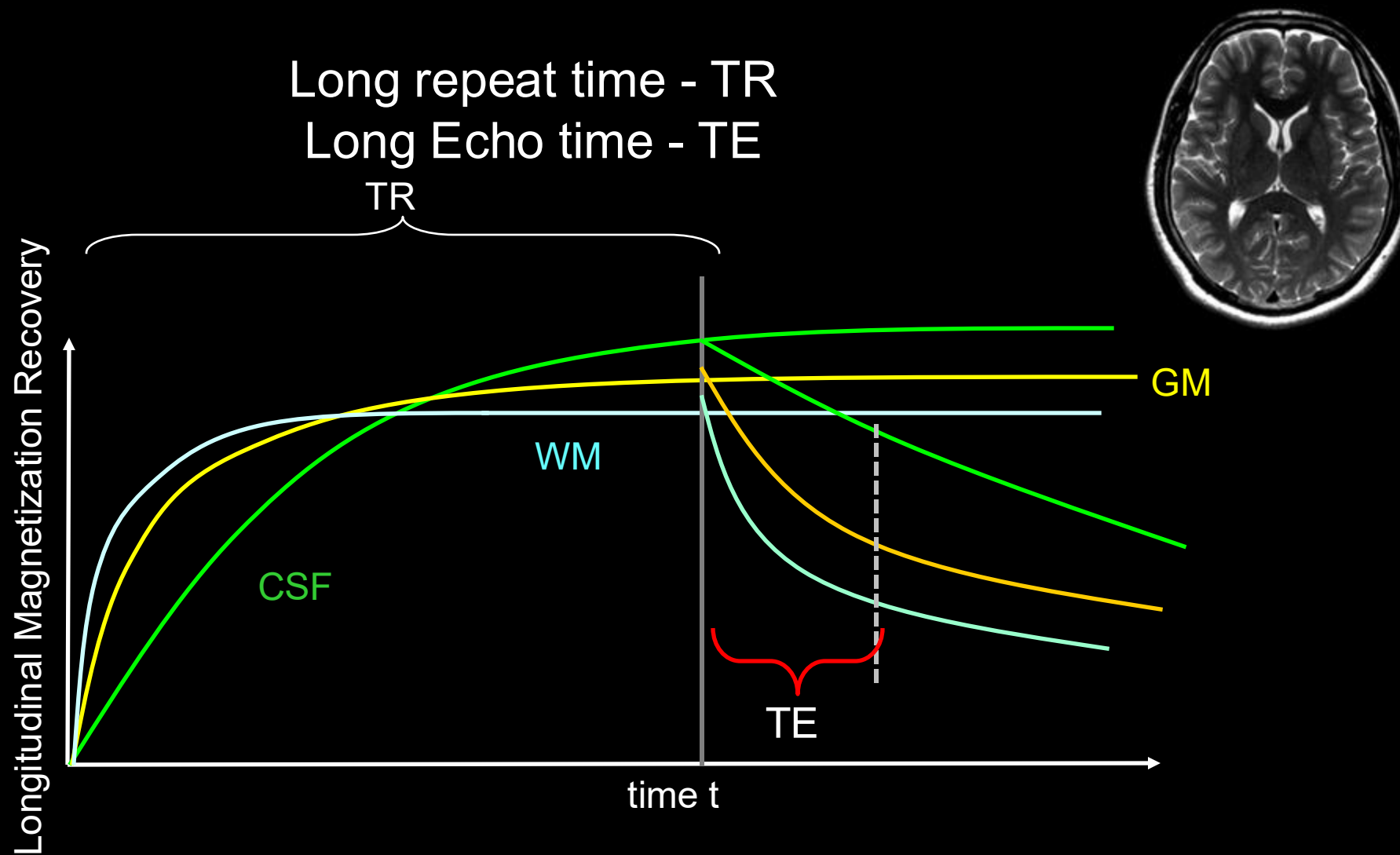
Short repetition time -TR
Short echo time - TE



Contrast

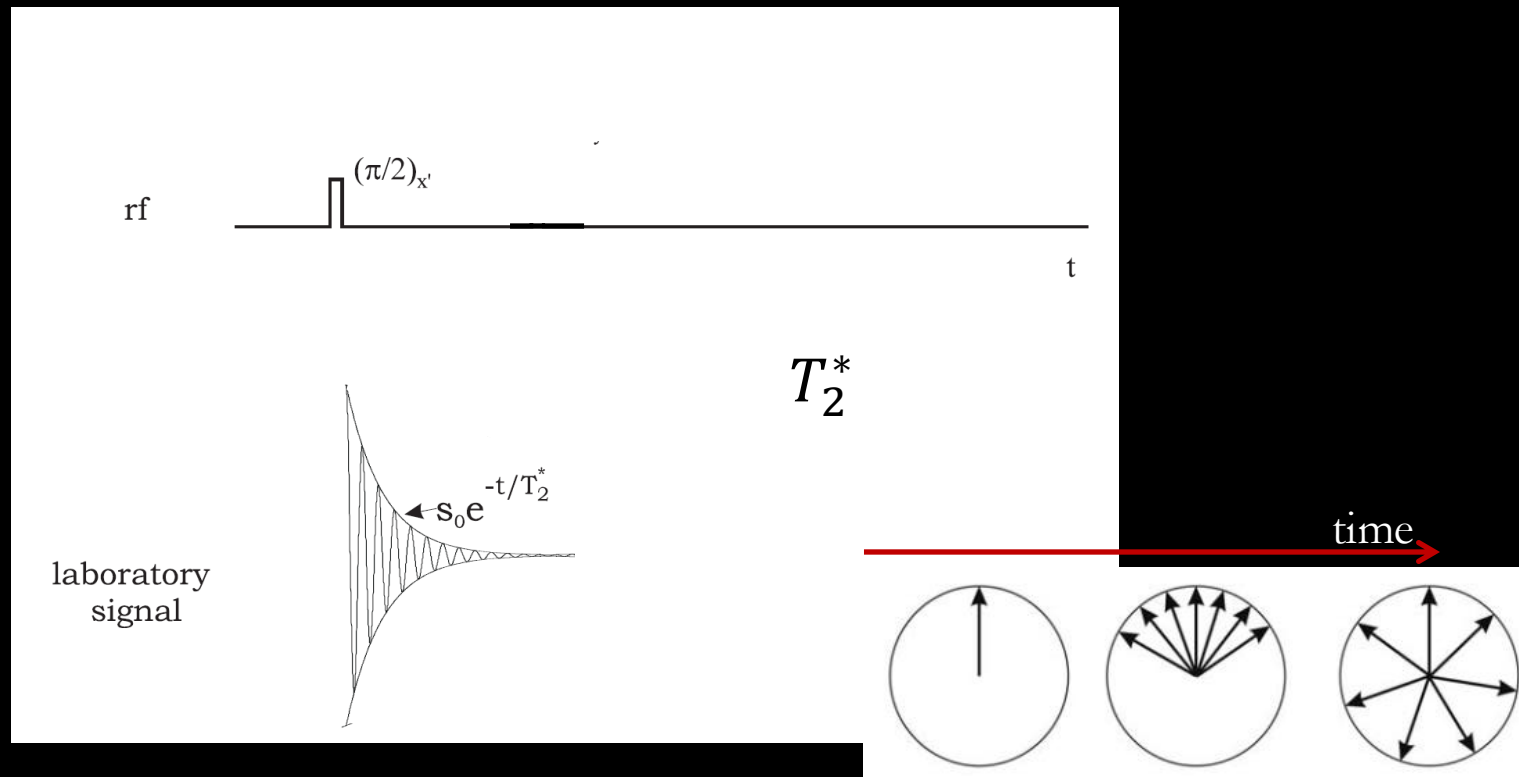


Contrast... T_2 weighting



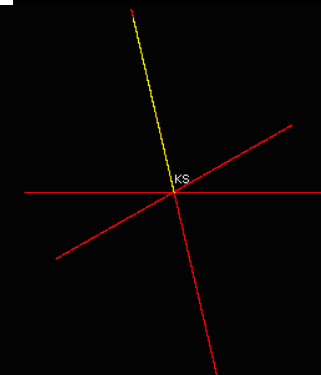
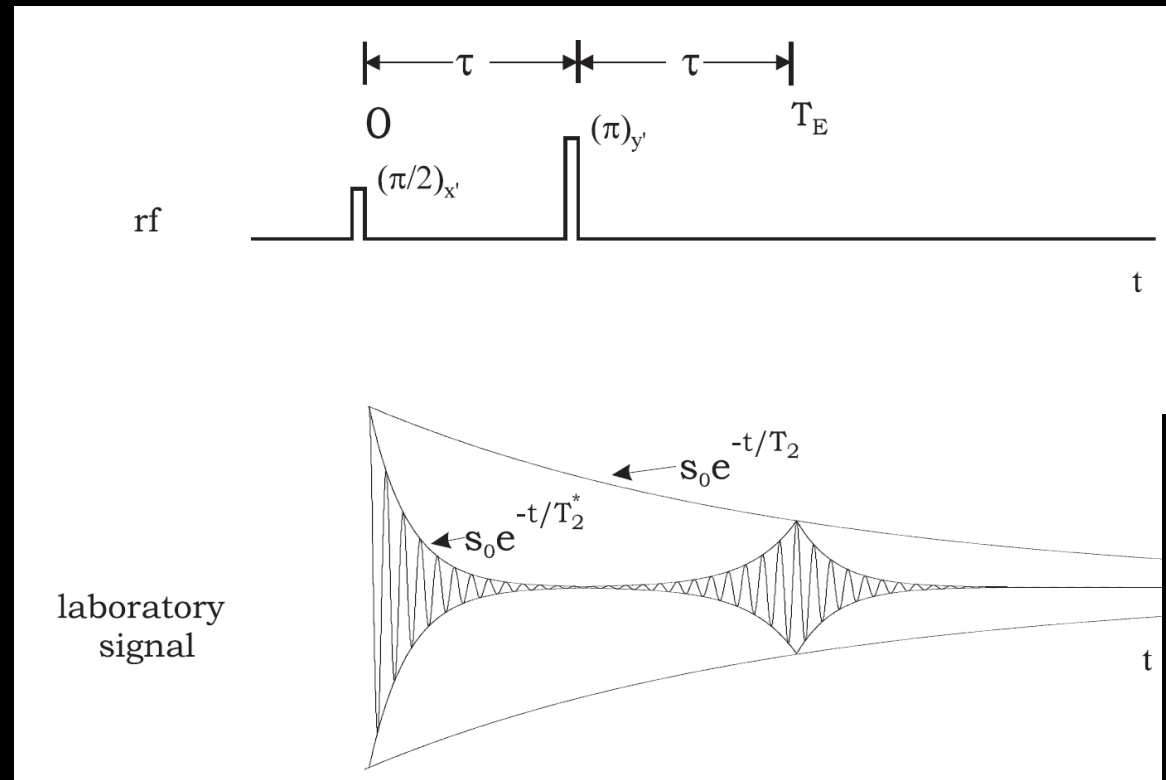
T_2 and T_2^* signal decay

Random fluctuations in the field experienced by protons, leads to incoherence and T_2 based signal decay

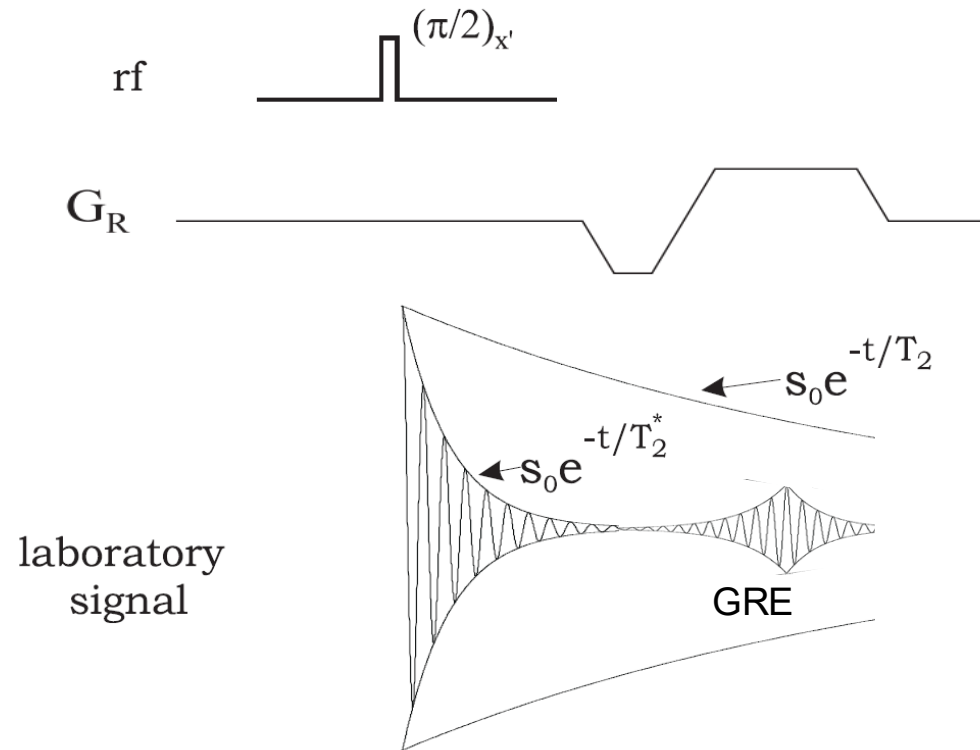


T_2 and T_2^* signal decay

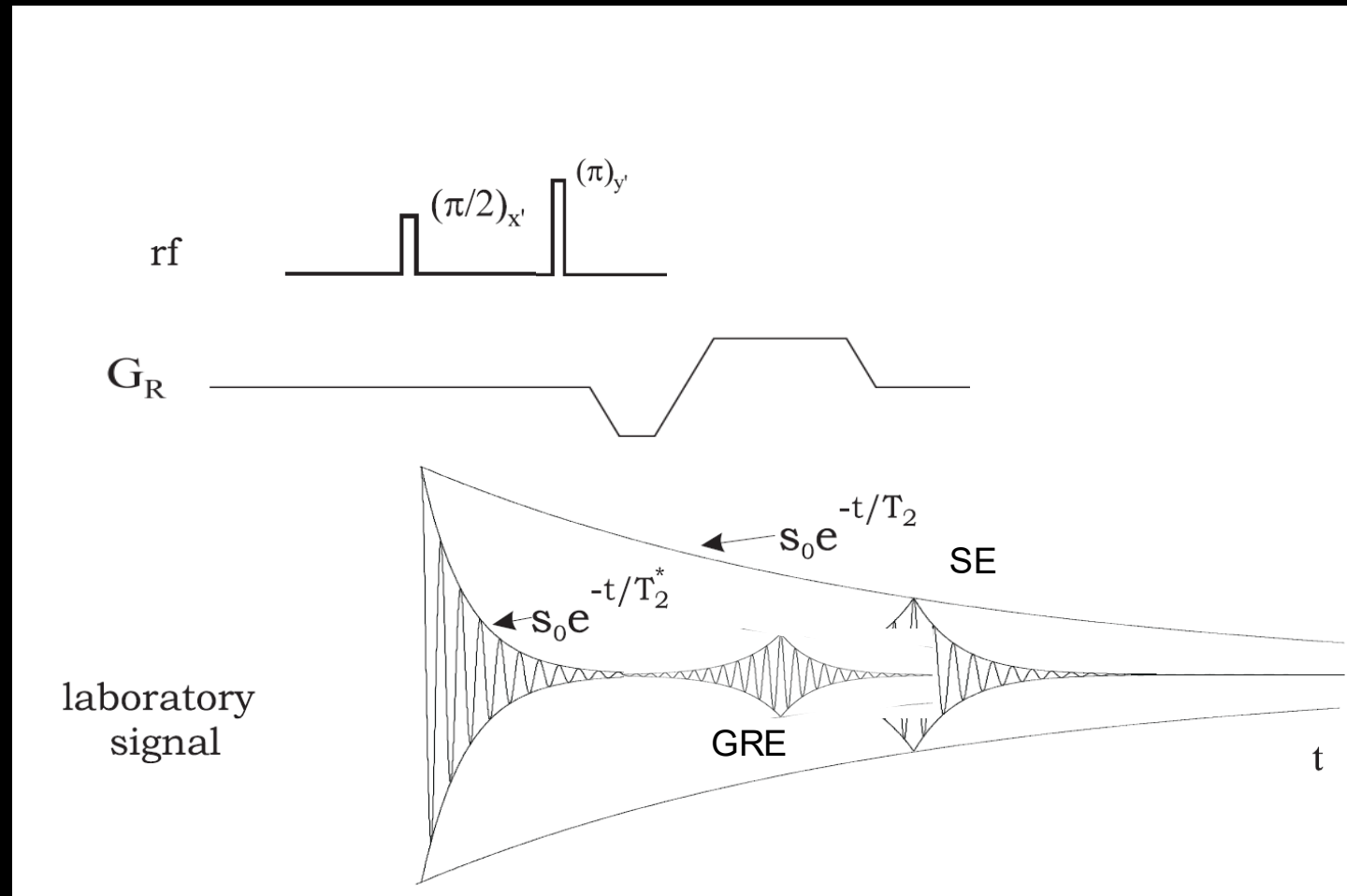
The field inhomogeneities that lead to T_2^* signal decay are deterministic in nature and hence are reversible with a spin echo



T2 and T2* signal decay: GRE vs. SE



T2 and T2* signal decay: GRE vs. SE



T2 and T2* signal decay: GRE vs. SE



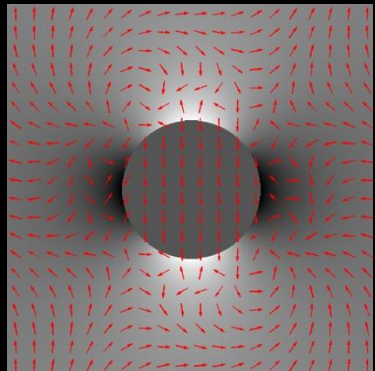
GE, T1W+T2*W



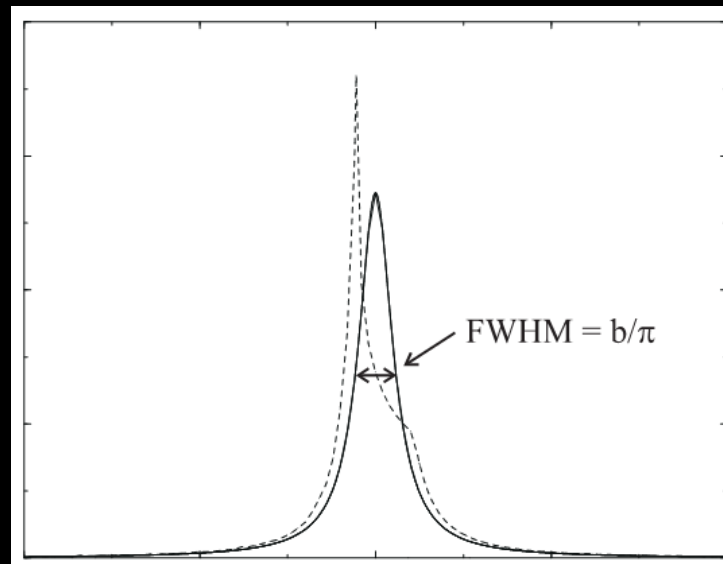
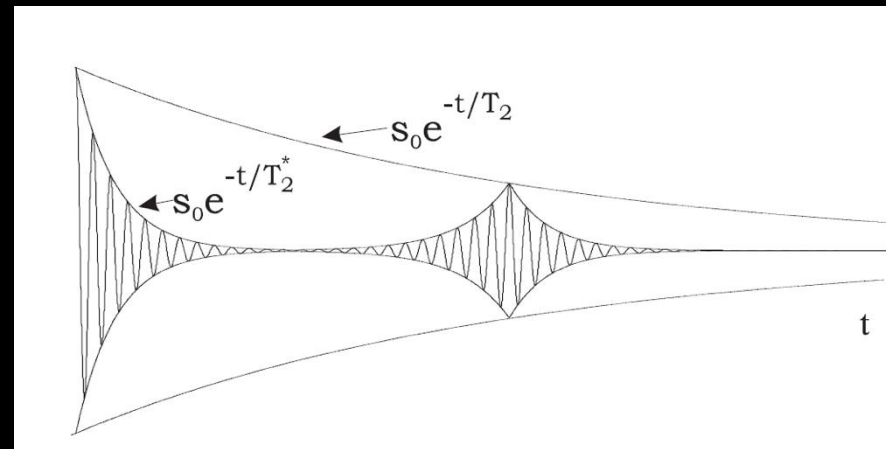
SE, T2W

T_2 and T_2^*

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_2'}$$

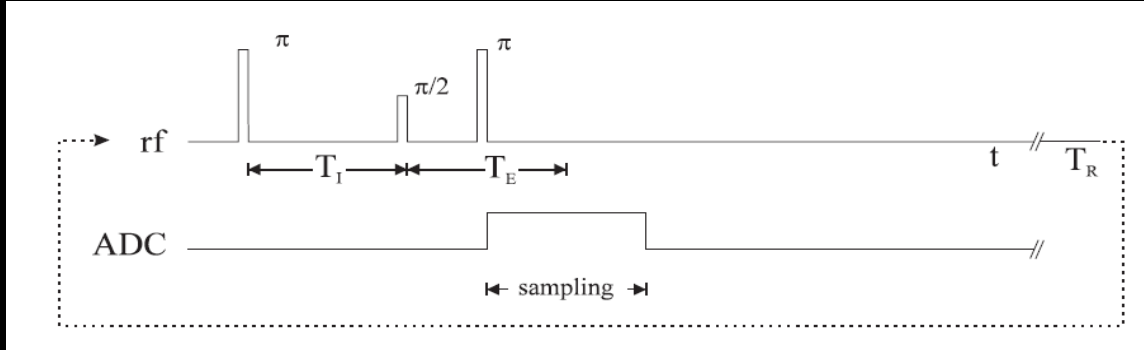


Histogram of distribution
of the resonance
frequencies, $\omega(r)$

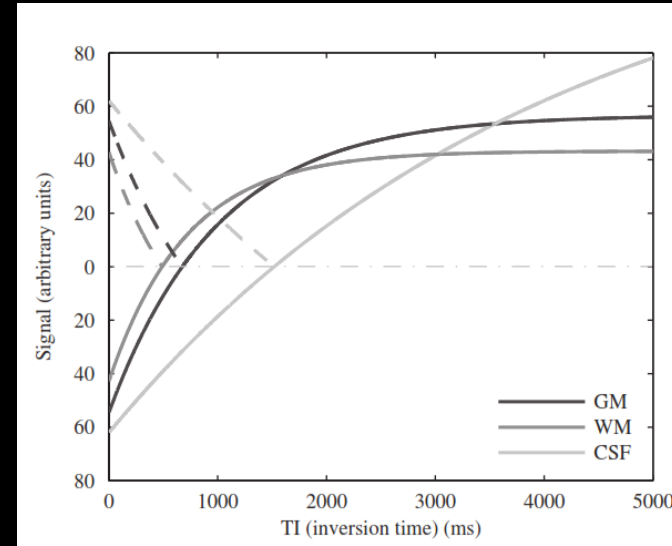


$$T_2' \simeq \frac{1}{|\gamma \cdot \Delta B|}$$

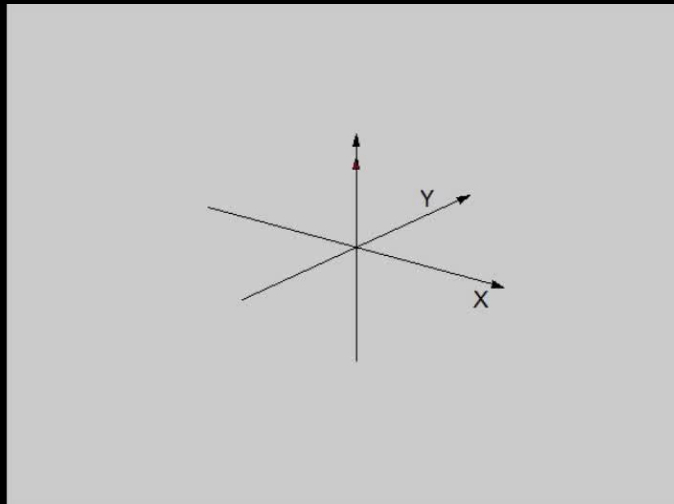
Inversion Recovery



An initial 180° degree pulse inverts the magnetization



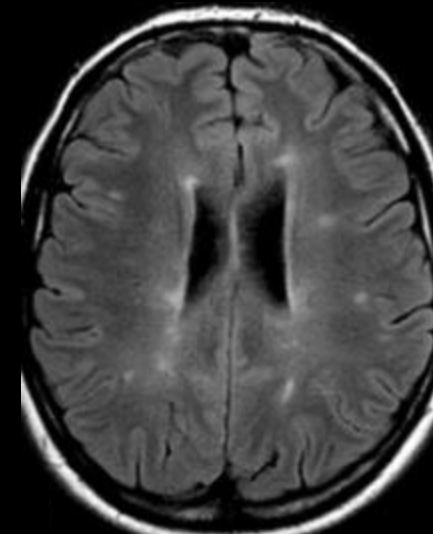
By appropriately timing (TI) the excitation and imaging sequence of pulses, signal from specific sequences can be nulled



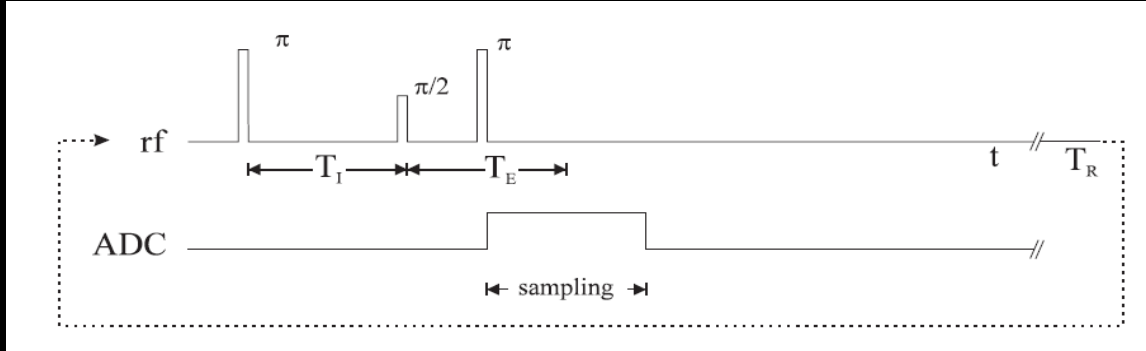
Appropriately Timed for signal nulling

FLAIR-

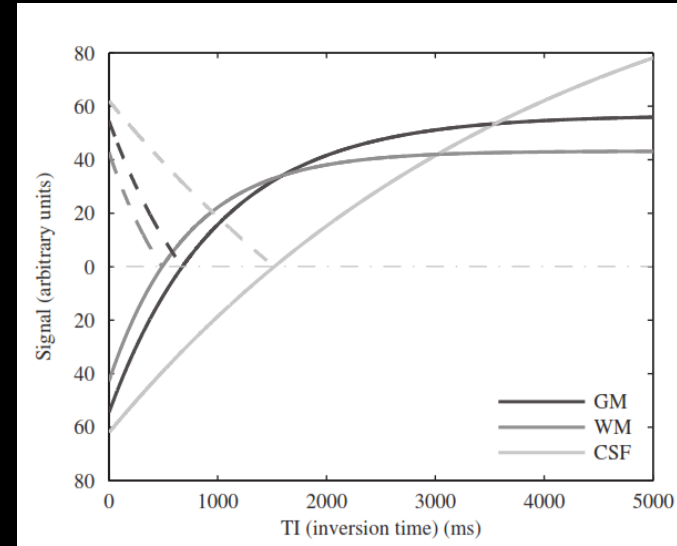
Fluid
attenuated
Inversion
recovery



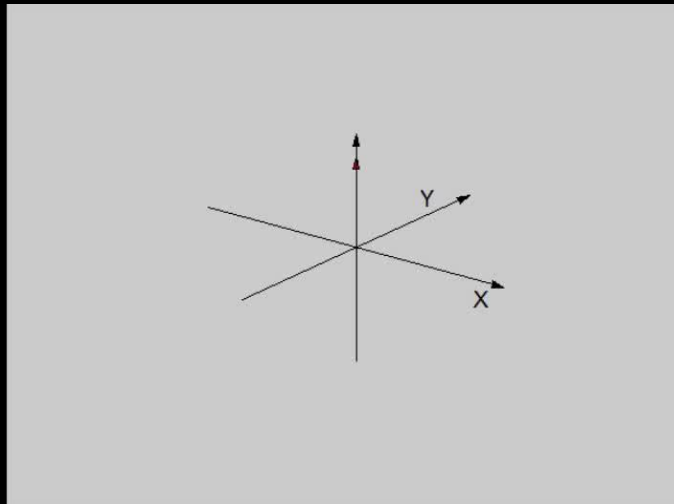
Inversion Recovery



An initial 180° degree pulse inverts the magnetization



By appropriately timing (TI) the excitation and imaging sequence of pulses, signal from specific sequences can be nulled



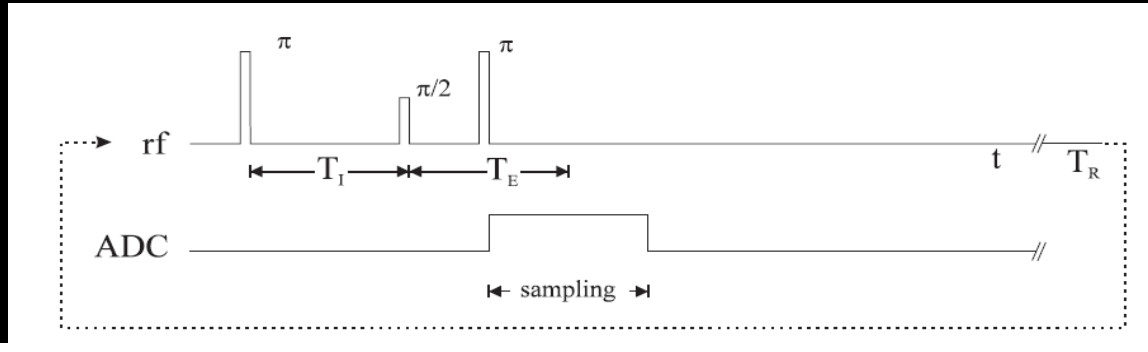
STIR

Fat is
saturated



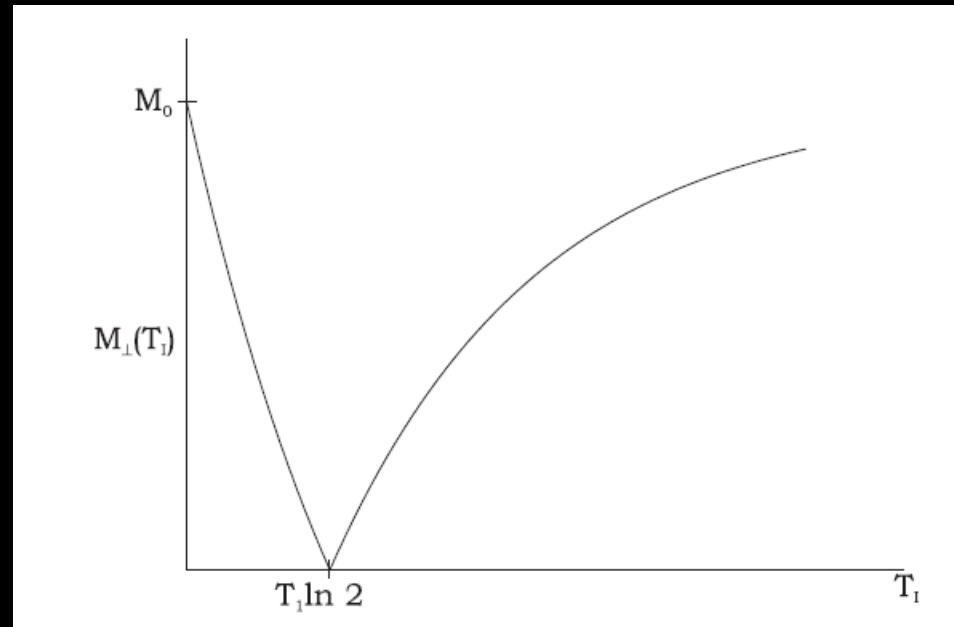
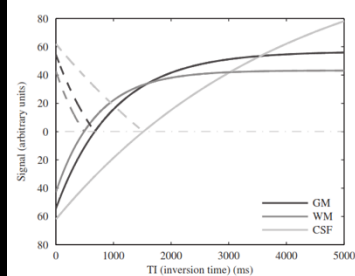
Appropriately Timed for signal nulling

Inversion Recovery

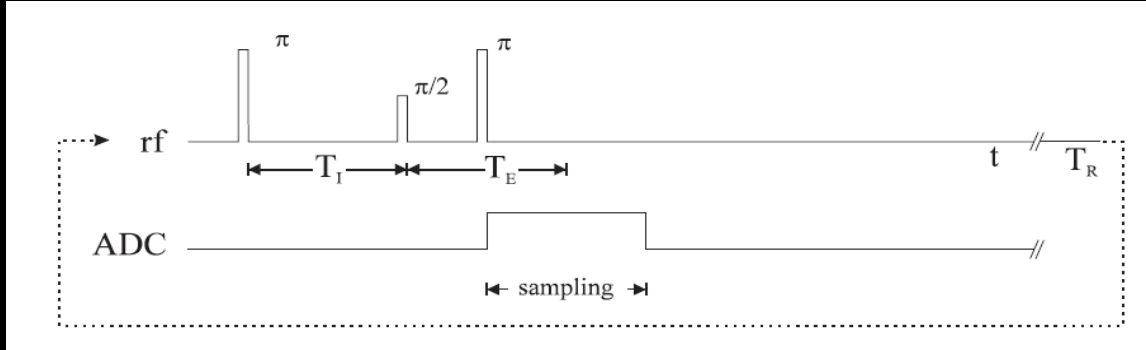


$$M_z(t) = M_0(1 - 2 \cdot e^{-\frac{t}{T_1}})$$

$$TI = T_1 \cdot \ln(2)$$

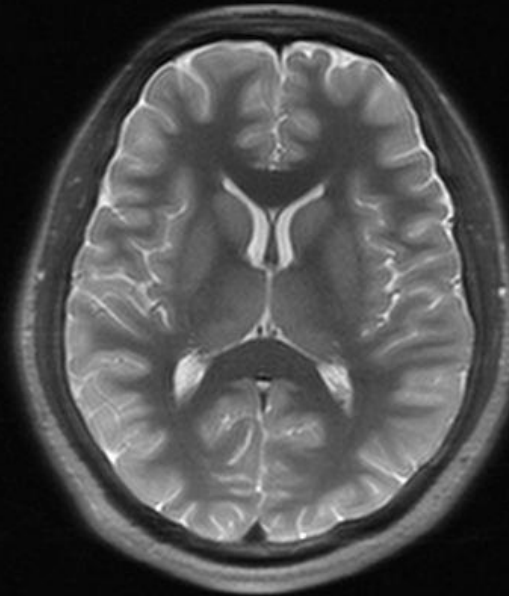


Inversion Recovery



$$M_z(t) = M_0(1 - 2 \cdot e^{-\frac{t}{T_1}})$$

$$TI = T_1 \cdot \ln(2)$$



Images acquired with different inversion times
(TIs from 100 to 1500 msec @1.5T)